a second patient, the genome of the RPE cell line was sequenced, and a mutation was discovered in a known oncogene. The trial was halted and a decision made to discontinue the effort for customized cell therapy in favor of using RPE cells derived from the national repository of banked iPSC lines which undergo extensive gene sequencing and quality controls. This outcome serves as a caution for the challenges involved in bringing a customized cell therapy to the clinic.

MESENCHYMAL STEM CELLS

By far the largest number of human trials have been performed using MSCs sourced from a variety of sites including bone marrow, peripheral blood, adipose tissue, umbilical cord, and other sites. Interest in the potential utility of MSCs for regenerative therapy began with the optimistic report that bone marrow stem cells were pluripotent and capable of generating nerve and heart muscle as well as blood cells. The possibility that easily obtainable MSCs could be used to regenerate injured or diseased cells or organs to treat diseases ranging from stroke, neurodegenerative disease, myocardial infarct, and even diabetes, generated enormous enthusiasm. The enthusiasm proved irresistible to many, and even after the initial reports were discredited—MSCs turned out not to be pluripotent stem cells as initially thought—a veritable flood of papers began to appear claiming disease-modifying activity of MSCs in mouse models of a wide range of degenerative disease and injury models. But when it became clear that the MSCs were not transforming into or generating new neurons or cardiac myocytes, alternative mechanisms of action were invoked, including the release of trophic factors, cytokines, or inflammatory modulators that were credited with producing their remarkable restorative effects. The relative ease with which blood or adipose tissue can be harvested from patients or donors and MSCs extracted has led to a rapidly expanding number of clinical trials for conditions ranging from stroke and MS to AD, ALS, and PD. Furthermore, a loophole in the regulatory framework of the FDA allows autologous cell therapy to escape regulation provided that the cells have not been significantly processed. This lax regulation has spawned a veritable industry of stem cell clinics making unsubstantiated claims of success in treating nervous system diseases. Patients have died from treatments in unregulated clinics operating in countries around the world, and three patients became blind in a well-publicized incident following stem cell treatments delivered by a Florida clinic. The "stem cells" were derived from the patients' own fat tissue and blood. These activities represent the dark side of the stem cell revolution perpetrated by practitioners who exploit the desperation of patients and their families. Legitimate and effective stem cell therapies will emerge over time, but given the prevalence and abundance of misleading information available on the Internet and elsewhere, a trusted and well-informed physician can play a key role in helping patients navigate the current cell therapy minefield.

MSCS FOR TRAUMATIC BRAIN INJURY

An allogeneic bone marrow–derived MSC line received conditional marketing approval in Japan in 2024 for the indication of improving chronic motor paralysis resulting from traumatic brain injury. The MSCs were transiently transfecting with the human Notch-1 intracellular domain gene to promote FDF-2 secretion in order to "enhance their ability to regenerate nerve cells" according to the pharmaceutical company that developed the cell-based therapy. The approval followed results of a phase 2 clinical trial conducted in Japan and the United States. Forty-six patients with moderate to severe traumatic brain injury and chronic motor deficits had MSCs stereotactically infused into an area of encephalomalacia identified on MRI scan while a sham group of 15 patients had burr holes only. The trial met the primary endpoint showing significant improvement in motor function at 24 weeks on the Fugle-Meyer Motor Scale (FMMS) ($p = .04$). Interestingly, a small improvement was noted in the sham-treated group as well, indicating the presence of a placebo effect. A larger, double-blind, randomized, sham-controlled study is now planned.

PERSPECTIVE

The premise that stem cell biology would herald an era of regenerative medicine has fueled exaggerated claims, false starts, and a proliferation of bogus clinics. But now we may be on the threshold of a new era of stem cell-based therapies for neurologic diseases and disorders including PD, spinal cord injury, ALS, and epilepsy. Whether this promise becomes reality will depend on the outcome of the first wave of pivotal double-blind controlled trials that are now being conducted or planned.

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Section 2 Diseases of the Central Nervous System

436 Seizures and Epilepsy

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A *seizure* (from the Latin *sacire*, "to take possession of") is a transient occurrence of signs or symptoms due to abnormal excessive or synchronous neuronal activity in the brain. Depending on the distribution of discharges, this abnormal brain activity can have various manifestations, ranging from dramatic convulsive activity to experiential phenomena not readily discernible by an observer. Although a variety of factors influence the incidence and prevalence of seizures, ~5–10% of the population will have at least one seizure, with the highest incidence occurring in early childhood and late adulthood.

The meaning of the term *seizure* needs to be carefully distinguished from that of epilepsy. *Epilepsy* describes a condition in which a person has a risk of *recurrent* seizures due to a chronic, underlying process. This definition implies that a person with a single seizure, or recurrent

seizures due to correctable or avoidable circumstances, does not necessarily have epilepsy (although a single seizure associated with clinical or electroencephalographic features portending high risk of recurrence may establish the diagnosis of epilepsy). Epilepsy refers to a clinical phenomenon rather than a single disease entity because many forms and causes exist. However, among the many causes of epilepsy, there are various *epilepsy syndromes* in which the clinical and pathologic characteristics are distinctive and suggest a specific underlying etiology.

Using the definition of epilepsy as two or more unprovoked seizures, the incidence of epilepsy is ~0.3–0.5% in different populations throughout the world, and the prevalence of epilepsy has been estimated at 5–30 persons per 1000.

CLASSIFICATION OF SEIZURES

Determining the type of seizure that has occurred is essential for focusing the diagnostic approach on particular etiologies, selecting appropriate therapy, and providing information regarding prognosis. The International League Against Epilepsy (ILAE) Commission on Classification and Terminology updated their approach to classification of seizures in 2017 **(Table 436-1)**. This system is based on the clinical features of seizures and associated electroencephalographic findings. Other potentially distinctive features such as etiology or cellular substrate are not considered in this classification system, although this will undoubtedly change in the future as more is learned about the pathophysiologic mechanisms that underlie specific seizure types.

A fundamental principle is that seizures may be either focal or generalized. *Focal seizures* originate within networks limited to one brain region (note that the term *partial seizures* is no longer used). *Generalized seizures* arise within and rapidly engage networks distributed across both cerebral hemispheres. Focal seizures are often associated with structural abnormalities of the brain. In contrast, generalized seizures may result from cellular, biochemical, or structural abnormalities with a more widespread distribution. There are exceptions in both cases, however.

FOCAL ONSET SEIZURES

Focal seizures arise from a neuronal network either discretely localized within one brain region or more broadly distributed but still within a cerebral hemisphere. With the new classification system, the subcategories of "simple focal seizures" and "complex focal seizures" have been eliminated. Instead, the classification emphasizes the effect on awareness (intact or impaired) and nature of the onset (motor or nonmotor). Focal seizures can also evolve into generalized seizures. In the past, this was referred to as *focal seizures with secondary generalization*, but the new system relies on descriptions of the type of generalized seizures that evolve from the focal seizure.

The routine interictal (i.e., between seizures) electroencephalogram (EEG) in patients with focal seizures is often normal or may show brief discharges termed *epileptiform spikes*, or *sharp waves*. Because focal seizures can arise from the medial temporal lobe or inferior frontal lobe (i.e., regions distant from the scalp), the EEG recorded during the seizure may be nonlocalizing. However, the region of seizure onset may be defined using surgically placed intracranial electrodes.

TABLE 436-1 Classification of Seizures[a]

1. **Focal Onset**
   (Can be further described as having intact or impaired awareness, motor or nonmotor onset, or evolve from focal to bilateral tonic clonic)
2. **Generalized Onset**
   a. Motor
      Tonic-clonic
      Other motor (e.g., atonic, myoclonic)
   b. Nonmotor (absence)
3. **Unknown Onset**
   Motor, nonmotor, or unclassified

[a]Based on the 2017 International League Against Epilepsy classification of seizure types. (Data from RS Fisher et al: Operational classification of seizure types by the International League Against Epilepsy: Position Paper of the ILAE Commission for Classification and Terminology. Epilepsia 58:522, 2017.)

Focal Seizures with Intact Awareness Focal seizures can have motor manifestations (such as tonic, clonic, or myoclonic movements) or nonmotor manifestations (such as sensory, autonomic, or emotional symptoms) without impairment of awareness. For example, a patient having a focal motor seizure arising from the right primary motor cortex near the area controlling hand movement will experience involuntary movements of the contralateral left hand. Since the cortical region controlling hand movement is immediately adjacent to the region for facial expression, the seizure may also cause abnormal movements of the face synchronous with the movements of the hand. The EEG recorded with scalp electrodes during the seizure (i.e., an ictal EEG) may show abnormal focal discharges over the corresponding area of cerebral cortex.

Three additional features of focal motor seizures are worth noting. First, in some patients, the abnormal motor movements may begin in a very restricted region, such as the fingers, and gradually progress (over seconds to minutes) to include a larger portion of the extremity. This phenomenon, described by Hughlings Jackson and known as a "Jacksonian march," represents the spread of seizure activity over a progressively larger region of motor cortex. Second, patients may experience a localized paresis (Todd's paralysis) for minutes to many hours in the involved region following the seizure. Third, in rare instances, the seizure may continue for hours or days. This condition, termed *epilepsia partialis continua*, is often refractory to medical therapy.

Focal seizures may also manifest as changes in somatic sensation (e.g., paresthesias), vision (flashing lights or formed hallucinations), equilibrium (sensation of falling or vertigo), or autonomic function (flushing, sweating, piloerection). Focal seizures arising from the temporal lobe may also cause the sensation of acrid odors (e.g., bleach or burning rubber) or tastes (e.g., bitter or metallic), or hearing sounds (simple noise or complex sounds), or an epigastric sensation that rises to the head. Some patients describe intense internal feelings such as fear, a dreamlike sense, depersonalization, déjá vu, or illusions that objects are growing smaller (micropsia) or larger (macropsia). These subjective, "experiential" events that are not directly observable by someone else are referred to as *auras*.

Focal Seizures with Impaired Awareness Focal seizures may also be accompanied by a transient impairment of the patient's ability to maintain normal contact with the environment. The patient is unable to respond appropriately to visual or verbal commands during the seizure and has impaired recollection or awareness of the ictal phase. The seizures frequently begin with an aura (i.e., a focal seizure without cognitive disturbance) that is stereotypic for the patient. The start of the ictal phase is often a motionless stare, which marks the onset of the period of impaired awareness. The impaired awareness may be accompanied by *automatisms*, involuntary movements that can involve basic behaviors, such as chewing, lip smacking, swallowing, or "picking" hand movements, or more elaborate behaviors, such as a display of emotion or running. The patient is typically disoriented following the seizure, and the transition to full recovery of consciousness may range from seconds to hours. Examination immediately following the seizure may show an anterograde amnesia or transient neurologic deficits (such as aphasia, hemi-neglect, or visual loss) caused by postictal inhibition of the cortical regions most involved in the seizure.

The range of potential clinical behaviors linked to focal seizures is so broad that extreme caution is advised before concluding that paroxysmal, stereotypic episodes of bizarre or atypical behavior are not due to seizure activity. In such cases, additional detailed EEG studies may be helpful.

EVOLUTION OF FOCAL SEIZURES TO GENERALIZED SEIZURES

Focal seizures can spread to involve both cerebral hemispheres and produce a generalized seizure, usually of the tonic-clonic variety (discussed below). This evolution is observed frequently following focal seizures arising from a region in the frontal lobe but may also be associated with focal seizures occurring elsewhere in the brain. A focal seizure that evolves into a generalized seizure is often difficult

to distinguish from a primary generalized onset tonic-clonic seizure because bystanders tend to emphasize the more dramatic, generalized convulsive phase of the seizure and overlook the more subtle, focal symptoms present at onset. In some cases, the focal onset of the seizure becomes apparent only when a careful history identifies a preceding aura. Often, however, the focal onset is not clinically evident and may be established only through careful EEG analysis. Nonetheless, distinguishing between these two entities is extremely important, because there are substantial differences in the evaluation and treatment of epilepsies characterized by focal versus generalized onset seizures.

GENERALIZED ONSET SEIZURES

Generalized seizures arise at some point in the brain but immediately and rapidly engage neuronal networks in both cerebral hemispheres. Several types of generalized seizures have features that place them in distinctive categories and facilitate clinical diagnosis.

Typical Absence Seizures Typical absence seizures are characterized by sudden, brief lapses of consciousness without loss of postural control. The seizure usually lasts for only seconds, consciousness returns as suddenly as it was lost, and there is no postictal confusion. Although the brief loss of consciousness may be clinically inapparent or the sole manifestation of the seizure, absence seizures are usually accompanied by subtle, bilateral motor signs such as rapid blinking, chewing movements, or small-amplitude, clonic movements of the hands.

Typical absence seizures are associated with a group of genetically determined epilepsies with onset usually in childhood (ages 4–10 years) or early adolescence and are the main seizure type in 15–20% of children with epilepsy. The seizures can occur hundreds of times per day, but the child may be unaware of or unable to convey their existence. Because the clinical signs of the seizures are subtle, especially to parents who may not have had previous experience with seizures, it is not surprising that the first clue to absence epilepsy is often unexplained "daydreaming" and a decline in school performance recognized by a teacher. Indeed, absence epilepsy is often misdiagnosed as an attention deficit disorder.

The electrophysiologic hallmark of typical absence seizures is a burst of generalized, symmetric, 3-Hz, spike-and-slow-wave discharges that begins and ends suddenly, superimposed on a normal EEG background. Periods of spike-and-slow-wave discharges lasting more than a few seconds usually correlate with clinical signs, but the EEG often shows many more brief bursts of abnormal cortical activity than suspected clinically. Hyperventilation tends to provoke these electrographic discharges and even the seizures themselves and is routinely used when recording the EEG.

Atypical Absence Seizures Atypical absence seizures have features that deviate both clinically and electrophysiologically from typical absence seizures. For example, the lapse of consciousness is usually longer and less abrupt in onset and cessation, and the seizure is accompanied by more obvious motor signs that may include focal or lateralizing features. The EEG shows a generalized, slow spike-and-slow-wave pattern with a frequency of ≤2.5 Hz, as well as other abnormal activity. Atypical absence seizures are usually associated with diffuse or multifocal structural abnormalities of the brain and therefore may accompany other signs of neurologic dysfunction, such as developmental delay. Furthermore, the seizures are less responsive to anticonvulsants compared to typical absence seizures.

Generalized, Tonic-Clonic Seizures Generalized onset tonic-clonic seizures are the main seizure type in ~10% of all persons with epilepsy. They are also the most common seizure type resulting from metabolic derangements and are therefore frequently encountered in many different clinical settings. The seizure usually begins abruptly without warning, although some patients describe vague premonitory symptoms in the hours leading up to the seizure. This prodrome is distinct from the stereotypic auras associated with focal seizures that generalize. The initial phase of the seizure is usually tonic contraction of muscles throughout the body, accounting for several classic features of the event. Tonic contraction of the muscles of expiration and the larynx at the onset will produce a loud moan or "ictal cry." Respirations are impaired, secretions pool in the oropharynx, and cyanosis develops. Contraction of the jaw muscles may cause biting of the tongue. A marked enhancement of sympathetic tone leads to increases in heart rate, blood pressure, and pupillary size. After 10–20 s, the tonic phase of the seizure typically evolves into the clonic phase, produced by the superimposition of periods of muscle relaxation on the tonic muscle contraction. The periods of relaxation progressively increase until the end of the ictal phase, which usually lasts no more than 1 min. The postictal phase is characterized by unresponsiveness, muscular flaccidity, and excessive salivation that can cause stridorous breathing and partial airway obstruction. Bladder or bowel incontinence may occur at this point. Patients gradually regain consciousness over minutes to hours, and, during this transition, there is typically a period of postictal confusion. Patients subsequently complain of headache, fatigue, and muscle ache that can last for many hours. The duration of impaired consciousness in the postictal phase can be extremely long (i.e., many hours) in patients with prolonged seizures or underlying central nervous system (CNS) disease.

The EEG during the tonic phase of the seizure shows a progressive increase in generalized low-voltage fast activity, followed by generalized high-amplitude, polyspike discharges. In the clonic phase, the high-amplitude activity is typically interrupted by slow waves to create a spike-and-slow-wave pattern. Generalized seizures tend to terminate synchronously over widespread brain regions. The postictal EEG shows diffuse suppression of all cerebral activity, followed by slowing that gradually recovers as the patient awakens.

There are several variants of generalized motor seizures, including pure tonic and pure clonic seizures. Brief tonic seizures lasting only a few seconds are especially noteworthy since they are usually associated with specific epilepsy syndromes having mixed seizure phenotypes, such as the Lennox-Gastaut syndrome (discussed below).

Atonic Seizures Atonic seizures are characterized by sudden loss of postural muscle tone lasting 1–2 s. Consciousness is briefly impaired, but there is usually no postictal confusion. A very brief seizure may cause only a quick head drop or nodding movement, whereas a longer seizure will cause the patient to collapse (hence, the less formal term, *drop attacks*). This can be extremely dangerous because there is a substantial risk of direct head injury with the fall. The EEG shows brief, generalized spike-and-wave discharges followed immediately by diffuse slow waves that correlate with the loss of muscle tone. Like pure tonic seizures, atonic seizures are usually seen in association with known epilepsy syndromes.

Myoclonic Seizures Myoclonus is a sudden and brief muscle contraction that may involve one part of the body or the entire body. A normal, common physiologic form of myoclonus is the sudden jerking movement observed while falling asleep. Pathologic myoclonus is most often seen in association with metabolic disorders, degenerative CNS diseases, or anoxic brain injury **(Chap. 318)**. Although the distinction from other forms of myoclonus is imprecise, myoclonic seizures are true epileptic events because they are caused by cortical (vs subcortical or spinal) dysfunction. The EEG shows bilaterally synchronous spike-and-slow-wave discharges immediately prior to the movement and muscle artifact associated with the myoclonus. Myoclonic seizures usually coexist with other forms of generalized seizures but are the predominant feature of juvenile myoclonic epilepsy (JME) (discussed below).

Epileptic Spasms Epileptic spasms are characterized by a briefly sustained flexion or extension of predominantly proximal muscles, including truncal muscles. The EEG usually shows *hypsarrhythmia*, a chaotic background of diffuse, large-amplitude slow waves and irregular, multifocal spikes and sharp waves. During the clinical spasm, there is a marked suppression of the EEG background (the "electrodecremental response"). The electromyogram (EMG) also reveals a characteristic rhomboid pattern that may help distinguish spasms from brief tonic and myoclonic seizures. Epileptic spasms occur predominantly

in infants and likely result from differences in neuronal function and connectivity in the immature versus mature CNS.

EPILEPSY SYNDROMES

Epilepsy syndromes are disorders in which epilepsy is a predominant feature, and there is sufficient evidence (e.g., through clinical, EEG, radiologic, or genetic observations) to suggest a common underlying mechanism. Three important epilepsy syndromes are listed below; additional examples with a known genetic basis are shown in **Table 436-2**.

JUVENILE MYOCLONIC EPILEPSY

JME is a generalized seizure disorder of unknown cause that appears in early adolescence and is usually characterized by bilateral myoclonic jerks that may be single or repetitive. The myoclonic seizures are most frequent in the morning after awakening and can be provoked by sleep deprivation. Awareness is preserved unless the myoclonus is especially severe. Many patients also experience generalized tonic-clonic seizures, and up to one-third have absence seizures. Although complete remission is uncommon, the seizures usually respond well to appropriate antiseizure medication. There is often a family history of epilepsy, and genetic studies suggest a polygenic cause.

LENNOX-GASTAUT SYNDROME

Lennox-Gastaut syndrome occurs in children and is defined by the following triad: (1) multiple seizure types (usually including generalized tonic-clonic, atonic, and atypical absence seizures); (2) an EEG with slow (<3 Hz) spike-and-wave discharges and a variety of other abnormalities; and (3) developmental delay. Lennox-Gastaut syndrome is associated with CNS disease or dysfunction from a variety of causes, including de novo mutations, developmental abnormalities, perinatal hypoxia/ischemia, trauma, infection, and other acquired lesions. The multifactorial nature of this syndrome suggests that it is a nonspecific response of the brain to diffuse neuronal dysfunction. Unfortunately, many patients have a poor prognosis due to the underlying CNS disease and the physical and psychosocial consequences of severe, poorly controlled epilepsy. Implanted neurostimulation devices (discussed below) are now being investigated for treatment of seizures in Lennox-Gastaut syndrome and other generalized epilepsies.

MESIAL TEMPORAL LOBE EPILEPSY SYNDROME

Mesial temporal lobe epilepsy (MTLE) is the most common syndrome associated with focal seizures with impairment of consciousness and is an example of an epilepsy syndrome with distinctive clinical, EEG, and pathologic features **(Table 436-3)**. High-resolution magnetic resonance imaging (MRI) can detect the characteristic hippocampal sclerosis that appears to be essential in the pathophysiology of MTLE for many patients **(Fig. 436-1)**. Recognition of this syndrome is especially important because it tends to be refractory to treatment with anticonvulsants but responds well to surgical intervention. Advances in the understanding of basic mechanisms of epilepsy have come through studies of experimental models of MTLE, discussed below.

THE CAUSES OF SEIZURES AND EPILEPSY

Seizures are a result of a shift in the normal balance of excitation and inhibition within the CNS. Given the numerous properties that control neuronal excitability, it is not surprising that there are many ways to perturb this normal balance and, therefore, many different causes of both seizures and epilepsy. Three clinical observations emphasize how a variety of factors determine why certain conditions may cause seizures or epilepsy in a given patient.

1. *The normal brain can have a seizure under the appropriate circumstances, and there are differences between individuals in the susceptibility or threshold for seizures*. For example, seizures may be induced by high fever in children who are otherwise normal and who never develop other neurologic problems, including epilepsy. However, febrile seizures occur only in a relatively small proportion of children. This implies there are various underlying *endogenous factors* that influence the threshold for having a seizure. Some of these factors are genetic, as a family history of epilepsy has a clear influence on the likelihood of seizures occurring in otherwise normal individuals. Normal development also plays an important role, because the brain appears to have different seizure thresholds at different maturational stages.
2. *There are a variety of conditions that have an extremely high likelihood of resulting in a chronic seizure disorder*. One of the best examples of this is severe, penetrating head trauma, which is associated with up to a 45% risk of subsequent epilepsy. The high propensity for severe traumatic brain injury to lead to epilepsy suggests that the injury results in a long-lasting pathologic change in the CNS that transforms a presumably normal neuronal network into one that is abnormally hyperexcitable. This process is known as *epileptogenesis*, and the specific changes that result in a lowered seizure threshold can be considered *epileptogenic factors*. Other processes associated with epileptogenesis include stroke, infection, neurodegeneration, and abnormalities of CNS development.
3. *Seizures are episodic*. Seizures occur intermittently, and, depending on the underlying cause, people with epilepsy may feel completely normal for months or years between seizures. This implies there are important provocative or *precipitating factors* that induce seizures in people with epilepsy. Similarly, precipitating factors are responsible for causing the single seizure in someone without epilepsy. Precipitants include those due to intrinsic physiologic processes, such as psychological or physical stress, sleep deprivation, or hormonal changes. They also include exogenous factors such as exposure to toxic substances, certain medications, and intermittent photic stimulation from strobe lights or some video games.

These observations emphasize the concept that the many causes of seizures and epilepsy result from a dynamic interplay between endogenous factors, epileptogenic factors, and precipitating factors. The potential role of each needs to be considered when determining the appropriate management of a patient with seizures. For example, the identification of predisposing factors (e.g., family history of epilepsy) in a patient with febrile seizures may increase the necessity for closer follow-up and a more aggressive diagnostic evaluation. Finding an epileptogenic lesion may help in the estimation of seizure recurrence and duration of therapy. Removal or modification of a precipitating factor may be an effective and safer method for preventing further seizures than the prophylactic use of anticonvulsant drugs. An emerging concept holds that underlying seizure risk itself fluctuates cyclically, potentially explaining why the same precipitating factor (e.g., sleep deprivation) can be well tolerated on some occasions but result in a seizure on others.

CAUSES ACCORDING TO AGE

In practice, it is useful to consider the etiologies of seizures based on the age of the patient, because age is one of the most important factors determining both the incidence and the likely causes of seizures or epilepsy **(Table 436-4)**. During the *neonatal period and early infancy*, potential causes include hypoxic-ischemic encephalopathy, trauma, CNS infection, congenital CNS abnormalities, and metabolic disorders. Babies born to mothers using neurotoxic drugs such as cocaine, heroin, or ethanol are susceptible to drug-withdrawal seizures in the first few days after delivery. Hypoglycemia and hypocalcemia, which can occur as secondary complications of perinatal injury, are also causes of early postnatal seizures. Seizures due to inborn errors of metabolism usually present once regular feeding begins, typically 2–3 days after birth. Pyridoxine (vitamin B_6) deficiency, an important cause of neonatal seizures, is treated with pyridoxine replacement. Idiopathic or familial forms of benign neonatal seizures are also seen during this time.

The most common seizures arising in *late infancy and early childhood* are febrile seizures, which are seizures associated with fevers but without evidence of CNS infection or other defined causes. The overall prevalence is 3–5% and even higher in some parts of the world such as Asia. Patients often have a family history of febrile seizures or epilepsy. Febrile seizures usually occur between 3 months and 5 years of age and have a peak incidence between 18 and 24 months. The typical scenario is a child who has a generalized, tonic-clonic seizure during a febrile illness in the setting of a common childhood infection such as otitis

TABLE 436-2 Examples of Genes Associated with Epilepsy Syndromes[a]

GENE (LOCUS)	FUNCTION OF GENE	CLINICAL SYNDROME	COMMENTS
CHRNA4 (20q13.2)	Nicotinic acetylcholine receptor subunit; mutations cause alterations in Ca^{2+} flux through the receptor; this may reduce the amount of GABA release in presynaptic terminals	Sleep-related hypermotor epilepsy (SHE); childhood onset; brief, nighttime seizures with prominent motor movements; often misdiagnosed as primary sleep disorder	Rare; first identified in a large Australian family; other families found to have mutations in *CHRNA2* or *CHRNB2*, and some families appear to have mutations at other loci
KCNQ2 (20q13.3)	Voltage-gated potassium channel subunits; mutation in pore regions may cause a 20–40% reduction of potassium currents, which will lead to impaired repolarization	Self-limited familial neonatal epilepsy; autosomal dominant inheritance; onset in first week of life in infants who are otherwise normal; remission usually within weeks to months; long-term epilepsy in 10–15%	Rare; other families found to have mutations in *KCNQ3*; sequence and functional homology to *KCNQ1*, mutations of which cause long QT syndrome and a cardiac-auditory syndrome
SCN1A (2q24.3)	α-Subunit of a voltage-gated sodium channel; numerous mutations affecting sodium currents that cause either gain or loss of function; network effects appear related to expression in excitatory or inhibitory cells	Very common cause of Dravet syndrome (severe myoclonic epilepsy of infancy) and some cases of Lennox-Gastaut syndrome. Also found in other syndromes, including genetic epilepsy with febrile seizures plus (GEFS+); autosomal dominant inheritance; presents with febrile seizures at median 1 year, which may persist >6 years, then variable seizure types not associated with fever	Incidence of Dravet syndrome is 1 in 20,000 births, and de novo *SCN1A* mutation is found in ~80% of cases. Incidence in GEFS+ uncertain; identified in other families with mutations in other sodium channel subunits (*SCN2B* and *SCN2A*) and $GABA_A$ receptor subunit (*GABRG2* and *GABRA1*); significant phenotypic heterogeneity within same family, including members with febrile seizures only. Avoid sodium channel–blocking antiseizure medications
LGI1 (10q24)	Leucine-rich glioma-inactivated 1 gene; previous evidence for role in glial tumor progression; recent studies suggest an influence in the postnatal development of glutamatergic circuits in the hippocampus	Autosomal dominant epilepsy with auditory features (ADEAF); a form of lateral temporal lobe epilepsy with auditory symptoms or aphasia as a major focal seizure manifestation; age of onset usually between 10 and 25 years	Mutations found in up to 50% of families containing two or more subjects with focal epilepsy with ictal auditory symptoms, suggesting that at least one other gene may underlie this syndrome
DEPDC5 (22q12.2)	Disheveled, Egl-10, and pleckstrin domain containing protein 5; exerts an inhibitory effect on mammalian target of rapamycin (mTOR)–mediated processes, such as cell growth and proliferation	Autosomal dominant familial focal epilepsy with variable foci (FFEVF); family members have seizures originating from different cortical regions; neuroimaging usually normal but may harbor subtle malformations; recent studies also suggest association with benign epilepsy with centrotemporal spikes	Study of families with the limited number of affected members revealed mutations in ~12% of families; thus, may be a relatively common cause of lesion-negative focal epilepsies with suspected genetic basis. Also associated with mutations in the GATOR1 genes *NPRL2* and *NPRL3*
GRIN2A (16p13.2)	Encodes NMDA receptor (NMDAR) subunit GluN2A	Spectrum of phenotypes ranging from benign childhood epilepsy with centrotemporal spikes (BECTS) to epilepsy-aphasia syndromes such as Landau-Kleffner syndrome (LKS) and other epileptic encephalopathies	Complex inheritance is implicated, and studies have shown considerable inter- and intrafamilial phenotypic variability and incomplete penetrance
CDKL-5 (Xp22.13)	Encodes cyclin-dependent kinase-like 5 (CDKL-5), a serine-threonine kinase involved in neural maturation and synaptogenesis	CDKL-5 deficiency disorder (CDD) results from pathogenic mutation in the *CDKL5* gene that causes absence or nonfunctional CDKL-5 protein. CDD is a severe developmental epileptic encephalopathy characterized by very-early-onset seizures. X-linked, affects females more than males	Ganaxolone is a recently approved antiseizure drug that has been shown to significantly reduce CDD-associated seizures
SLC2A1 (1p34.2)	Glucose transporter protein type 1 (GLUT1); transports glucose across the blood-brain barrier	Loss of function of one allele leads to GLUT1 deficiency, a severe metabolic encephalopathy including intractable epilepsy, complex motor dysfunction, and intellectual disability. Milder GLUT1 deficiency causes a combination of movement disorder (paroxysmal exertional dyskinesia) and epilepsy with prominent absence seizures, though intellect is often normal	Milder forms of epilepsy due to GLUT1 deficiency may respond to standard antiseizure medications, but the gold standard treatment for refractory forms is the ketogenic diet, which bypasses defective glucose transport to provide an alternative energy supply to the brain
CSTB (21q22.3)	Cystatin B, a noncaspase cysteine protease inhibitor; normal protein may block neuronal apoptosis by inhibiting caspases directly or indirectly (via cathepsins), or controlling proteolysis	Progressive myoclonus epilepsy (PME) (Unverricht-Lundborg disease); autosomal recessive inheritance; age of onset between 6 and 15 years, myoclonic seizures, ataxia, and progressive cognitive decline; brain shows neuronal degeneration	Overall rare, but relatively common in Finland and western Mediterranean (>1 in 20,000); precise role of cystatin B in human disease unknown, although mice with null mutations of cystatin B have similar syndrome
EPM2A (6q24)	Laforin, a protein tyrosine phosphatase (PTP); involved in glycogen metabolism and may have antiapoptotic activity	Progressive myoclonus epilepsy (Lafora's disease); autosomal recessive inheritance; age of onset 6–19 years, death within 10 years; brain degeneration associated with polyglucosan intracellular inclusion bodies in numerous organs	Most common PME in southern Europe, Middle East, northern Africa, and Indian subcontinent; genetic heterogeneity; unknown whether seizure phenotype due to degeneration or direct effects of abnormal laforin expression
Doublecortin (Xq21-24)	Doublecortin, expressed primarily in frontal lobes; directly regulates microtubule polymerization and bundling	Classic lissencephaly associated with severe mental retardation and seizures in males; subcortical band heterotopia with more subtle findings in females (presumably due to random X inactivation); X-linked dominant	Relatively rare but of uncertain incidence; recent increased ascertainment due to improved imaging techniques; relationship between migration defect and seizure phenotype unknown

[a]The first seven syndromes listed in the table (SHE, benign familial neonatal convulsions, GEFS+, ADEAF, FFEVF, BECTS, LKS, CDD) are examples of genetic epilepsies associated with identified gene mutations. The last three syndromes are examples of the numerous Mendelian disorders in which seizures are one part of the phenotype.

Abbreviation: GABA, γ-aminobutyric acid.

TABLE 436-3 Characteristics of the Mesial Temporal Lobe Epilepsy Syndrome	
History	
History of febrile seizures	Rare generalized seizures
Family history of epilepsy	Seizures may remit and reappear
Early onset	Seizures often intractable
Clinical Observations	
Aura common	Postictal disorientation
Behavioral arrest/stare	Memory loss
Complex automatisms	Dysphasia (with focus in dominant hemisphere)
Unilateral posturing	
Laboratory Studies	
Unilateral or bilateral anterior temporal spikes on EEG	
Hypometabolism on interictal PET	
Hyperperfusion on ictal SPECT	
Material-specific memory deficits on intracarotid amobarbital (Wada) test	
MRI Findings	
Small hippocampus with increased signal on T2-weighted sequences and loss of trilaminar hippocampal internal architecture	
Small temporal lobe	
Enlarged temporal horn	
Pathologic Findings	
Highly selective loss of specific cell populations within hippocampus in most cases, granule cell layer dispersion, gliosis	

Abbreviations: EEG, electroencephalogram; MRI, magnetic resonance imaging; PET, positron emission tomography; SPECT, single-photon emission computed tomography.

media, respiratory infection, or gastroenteritis. The seizure is likely to occur during the rising phase of the temperature curve (i.e., during the first day) rather than well into the course of the illness. A *simple* febrile seizure is a single, isolated event, brief, and symmetric in appearance. *Complex* febrile seizures are characterized by repeated seizure activity, a duration >15 minutes, or by focal features. Approximately one-third of patients with febrile seizures will have a recurrence, but <10% have three or more episodes. Recurrences are much more likely when the febrile seizure occurs in the first year of life. Simple febrile seizures are not associated with increased risk of developing epilepsy, while complex febrile seizures have a risk of 2–5%; other risk factors include the presence of preexisting neurologic deficits and a family history of nonfebrile seizures.

FIGURE 436-1 Mesial temporal lobe epilepsy. The electroencephalogram and seizure semiology were consistent with a left temporal lobe focus. This coronal high-resolution T2-weighted fast spin echo magnetic resonance image obtained at 3 Tesla is at the level of the hippocampal bodies and shows abnormal high signal intensity, blurring of internal laminar architecture, and reduced size of the left hippocampus (arrow) relative to the right. This triad of imaging findings is consistent with hippocampal sclerosis.

TABLE 436-4 Causes of Seizures	
Neonates (<1 month)	Perinatal hypoxia and ischemia Intracranial hemorrhage and trauma CNS infection Metabolic disturbances (hypoglycemia, hypocalcemia, hypomagnesemia, pyridoxine deficiency) Drug withdrawal Developmental disorders Genetic disorders
Infants and children (>1 month and <12 years)	Febrile seizures Genetic disorders (metabolic, degenerative, primary epilepsy syndromes) CNS infection Developmental disorders Trauma
Adolescents (12–18 years)	Trauma Genetic disorders Infection Illicit drug use Brain tumor
Young adults (18–35 years)	Trauma Alcohol withdrawal Illicit drug use Brain tumor Autoantibodies
Older adults (>35 years)	Cerebrovascular disease Brain tumor Alcohol withdrawal Metabolic disorders (uremia, hepatic failure, electrolyte abnormalities, hypoglycemia, hyperglycemia) Alzheimer's disease and other degenerative CNS diseases Autoantibodies

Abbreviation: CNS, central nervous system.

Childhood marks the age at which many of the well-defined epilepsy syndromes present. Some children who are otherwise normal develop idiopathic, generalized tonic-clonic seizures without other features that fit into specific syndromes. Temporal lobe epilepsy usually presents in childhood and may be related to mesial temporal lobe sclerosis (as part of the MTLE syndrome) or other focal abnormalities, such as cortical dysgenesis. Other types of focal seizures, including those that evolve into generalized seizures, may be late manifestations of a developmental disorder, an acquired lesion such as head trauma, CNS infection (e.g., viral encephalitis), or a CNS tumor.

The period of *adolescence and early adulthood* is one of transition during which the idiopathic or genetically based epilepsy syndromes, including JME and juvenile absence epilepsy, become less common, while epilepsies secondary to acquired CNS lesions begin to predominate. Seizures that arise in patients in this age range may be associated with head trauma, CNS infections (including parasitic infections such as cysticercosis), brain tumors, congenital CNS abnormalities, illicit drug use, or alcohol withdrawal. Autoantibodies directed against CNS antigens such as potassium channels or glutamate receptors are a cause of epilepsy that also begins to appear in this age group (although cases of autoimmunity are being increasingly described in the pediatric population), including patients without an identifiable cancer. This etiology should be suspected when a previously normal individual presents

with fulminant seizures, especially when combined with psychiatric symptoms and changes in cognitive function.

Head trauma is a common cause of epilepsy in adolescents and adults. The head injury can be caused by a variety of mechanisms, and the likelihood of developing epilepsy is strongly correlated with the severity of the injury. A patient with a penetrating head wound, depressed skull fracture, intracranial hemorrhage, or prolonged posttraumatic coma or amnesia has a 30–50% risk of developing epilepsy, whereas a patient with a closed head injury and cerebral contusion has a 5–25% risk. Recurrent seizures usually develop within 1 year after head trauma, although intervals of >10 years are well known. In controlled studies, mild head injury, defined as a concussion with amnesia or loss of consciousness of <30 min, was found to be associated with only a slightly increased likelihood of epilepsy. Nonetheless, some patients experience focal seizures within hours or days of a mild head injury and subsequently develop chronic seizures of the same type; such cases may represent rare examples of epilepsy resulting from mild head injury.

The causes of seizures in *older adults* include cerebrovascular disease, trauma (including subdural hematoma), CNS tumors, and degenerative diseases. Cerebrovascular disease may account for ~50% of new cases of epilepsy in patients >65 years. Acute poststroke seizures (i.e., occurring within the first 24 h after acute stroke) are more common after hemorrhagic strokes compared to ischemic strokes. Chronic seizures typically appear months to years after the initial event and are associated with all forms of stroke.

Metabolic disturbances such as electrolyte imbalance, hypo- or hyperglycemia, renal failure, and hepatic failure may cause seizures at any age. Similarly, endocrine disorders, hematologic disorders, vasculitides, and many other systemic diseases may cause seizures over a broad age range. Many medications and abused substances can precipitate seizures as well (Table 436-5).

TABLE 436-5 Drugs and Other Substances That Can Cause Seizures

Alkylating agents (e.g., busulfan, chlorambucil)
Antimalarials (chloroquine, mefloquine)
Antimicrobials/antivirals
β-Lactam and related compounds
Quinolones
Acyclovir
Isoniazid
Ganciclovir
Anesthetics and analgesics
Meperidine
Fentanyl
Tramadol
Local anesthetics
Dietary supplements
Ephedra (ma huang)
Gingko
Immunomodulatory drugs
Cyclosporine
OKT3 (monoclonal antibodies to T cells)
Tacrolimus
Interferons
Psychotropics
Antidepressants (e.g., bupropion)
Antipsychotics (e.g., clozapine)
Lithium
Radiographic contrast agents
Drug withdrawal
Alcohol
Baclofen
Barbiturates (short-acting)
Benzodiazepines (short-acting)
Zolpidem
Drugs of abuse
Amphetamine
Cocaine
Phencyclidine
Methylphenidate
Flumazenil[a]

[a]In benzodiazepine-dependent patients.

BASIC MECHANISMS

MECHANISMS OF SEIZURE INITIATION AND PROPAGATION

Focal seizure activity can begin in a discrete region of cortex and then slowly invade the surrounding regions. The hallmark of an established seizure is typically an electrographic "spike" due to intense near-simultaneous firing of many local excitatory neurons, resulting in an apparent hypersynchronization of the excitatory bursts across a relatively large cortical region. The bursting activity in individual neurons (the "paroxysmal depolarization shift") is caused by a relatively long-lasting depolarization of the neuronal membrane due to influx of extracellular calcium (Ca^{2+}), which leads to the opening of voltage-dependent sodium (Na^+) channels, influx of Na^+, and generation of repetitive action potentials. This is followed by a hyperpolarizing afterpotential mediated by γ-aminobutyric acid (GABA) receptors or potassium (K^+) channels, depending on the cell type. The synchronized bursts from enough neurons result in summation of field potentials producing a spike discharge on the EEG.

The spreading seizure wavefront is thought to slow and ultimately halt by intact hyperpolarization and a "surround" inhibition created by feedforward activation of inhibitory neurons. With sufficient activation, there is a recruitment of surrounding neurons via a number of synaptic and nonsynaptic mechanisms, including (1) an increase in extracellular K^+, which blunts hyperpolarization and depolarizes neighboring neurons; (2) accumulation of Ca^{2+} in presynaptic terminals, leading to enhanced neurotransmitter release; (3) depolarization-induced activation of the *N*-methyl-D-aspartate (NMDA) subtype of the excitatory amino acid receptor, which causes additional Ca^{2+} influx and neuronal activation; and (4) ephaptic interactions related to changes in tissue osmolarity and cell swelling. The recruitment of a sufficient number of neurons leads to the propagation of excitatory currents into contiguous areas via local cortical connections and to more distant areas via long commissural pathways such as the corpus callosum.

Many factors control neuronal excitability, and thus, there are many potential mechanisms for altering a neuron's propensity to have bursting activity. Mechanisms *intrinsic* to the neuron include changes in the conductance of ion channels, response characteristics of membrane receptors, cytoplasmic buffering, second-messenger systems, and protein expression as determined by gene transcription, translation, and posttranslational modification. Mechanisms *extrinsic* to the neuron include changes in the amount or type of neurotransmitters present at the synapse, modulation of receptors by extracellular ions and other molecules, and temporal and spatial properties of synaptic and nonsynaptic input. Glial cells, such as astrocytes and oligodendrocytes, have an important role in many of these mechanisms as well.

Certain recognized causes of seizures are explained by these mechanisms. For example, accidental ingestion of domoic acid, an analogue of glutamate (the principal excitatory neurotransmitter in the brain) produced by naturally occurring microscopic algae, causes profound seizures via direct activation of excitatory amino acid receptors throughout the CNS. Penicillin, which can lower the seizure threshold in humans and is a potent convulsant in experimental models, reduces inhibition by antagonizing the effects of GABA at its receptor. The basic mechanisms of other precipitating factors of seizures, such as sleep deprivation, fever, alcohol withdrawal, hypoxia, and infection, are not as well understood but presumably involve analogous perturbations in

neuronal excitability. Similarly, the endogenous factors that determine an individual's seizure threshold may relate to these properties as well.

Knowledge of the mechanisms responsible for initiation and propagation of most generalized seizures (including tonic-clonic, myoclonic, and atonic types) remains rudimentary and reflects the limited understanding of the connectivity of the brain at a systems level. Much more is understood about the origin of generalized spike-and-wave discharges in absence seizures. These appear to be related to oscillatory rhythms normally generated during sleep by circuits connecting the thalamus and cortex. This oscillatory behavior involves an interaction between $GABA_B$ receptors, T-type Ca^{2+} channels, and K^+ channels located within the thalamus. Pharmacologic studies indicate that modulation of these receptors and channels can induce absence seizures, and there is good evidence that the genetic forms of absence epilepsy may be associated with mutations of components of this system.

MECHANISMS OF EPILEPTOGENESIS

Epileptogenesis refers to the transformation of a normal neuronal network into one that is chronically hyperexcitable. There is often a delay of months to years between an initial CNS injury such as trauma, stroke, or infection and the first clinically evident seizure. The injury appears to initiate a process that gradually lowers the seizure threshold in the affected region until a spontaneous seizure occurs. In many genetic and idiopathic forms of epilepsy, epileptogenesis is presumably determined by developmentally regulated events.

Pathologic studies of the hippocampus from patients with temporal lobe epilepsy suggest that some forms of epileptogenesis are related to *structural changes in neuronal networks*. For example, many patients with MTLE have a highly selective loss of neurons that normally contribute to inhibition of the main excitatory neurons within the dentate gyrus. There is also evidence that, in response to the loss of neurons, there is reorganization of surviving neurons in a way that affects the excitability of the network. Some of these changes can be seen in experimental models of prolonged electrical seizures or traumatic brain injury. Thus, an initial injury such as head injury may lead to a focal region of structural change that causes local hyperexcitability. The local hyperexcitability leads to further structural changes that evolve over time until the focal lesion produces clinically evident seizures. Similar models have provided strong evidence for long-term alterations in *intrinsic, biochemical properties of cells* within the network such as chronic changes in glutamate or GABA receptor function. Induction of inflammatory cascades may be a critical factor in these processes as well.

GENETIC CAUSES OF EPILEPSY

An area of ongoing progress in epilepsy research has been the identification of genetic mutations associated with a variety of epilepsy syndromes (Table 436-2). Although most of the mutations identified to date cause rare forms of epilepsy, their discovery has led to extremely important conceptual advances. For example, it appears that many of the inherited epilepsies are due to mutations affecting ion channel function. These syndromes are therefore part of the larger group of channelopathies causing paroxysmal disorders such as cardiac arrhythmias, episodic ataxia, periodic weakness, and familial hemiplegic migraine. Other gene mutations are proving to be associated with pathways influencing CNS development, synaptic physiology, or neuronal homeostasis. De novo and somatic mutations may explain a significant proportion of these syndromes, especially those with onset in early childhood. A current challenge is to identify the multiple susceptibility genes that underlie the more common forms of idiopathic epilepsies. Ion channel mutations and copy number variants may be pathogenic in a subset of these patients.

MECHANISMS OF ACTION OF ANTISEIZURE DRUGS

Antiseizure drugs appear to act primarily by blocking the initiation or spread of seizures. This occurs through a variety of mechanisms that modify the activity of ion channels or neurotransmitters, and in most cases, the drugs have pleiotropic effects. The mechanisms include inhibition of Na^+-dependent action potentials in a frequency-dependent manner (e.g., phenytoin, carbamazepine, lamotrigine, topiramate, zonisamide, lacosamide, rufinamide, cenobamate), inhibition of voltage-gated Ca^{2+} channels (phenytoin, gabapentin, pregabalin), facilitating the opening of potassium channels (ezogabine), attenuation of glutamate activity (lamotrigine, topiramate, felbamate, perampanel), potentiation of GABA receptor function (benzodiazepines, barbiturates), increase in the availability of GABA (valproic acid, gabapentin, tiagabine), and modulation of release of synaptic vesicles (levetiracetam, brivaracetam). Two of the effective drugs for absence seizures, ethosuximide and valproic acid, probably act by inhibiting T-type Ca^{2+} channels in thalamic neurons. Cannabidiol (CBD), a derivative of cannabis plants, is effective for reducing seizures in children with Dravet syndrome, Lennox-Gastaut syndrome, and tuberous sclerosis but does not act through endogenous cannabinoid receptors. Rather, CBD has a multimodal mechanism of action involving modulation of intracellular calcium via G protein–coupled receptor 55, extracellular calcium influx via transient receptor potential vanilloid type 1 (TRPV1) channels, and adenosine-mediated signaling. Fenfluramine is an amphetamine derivative that probably exerts its antiseizure effect by inhibiting serotonin reuptake and increasing extracellular levels. Ganaxolone is a synthetic analogue of allopregnanolone and an allosteric modulator of $GABA_A$ receptors that reduces seizures associated with cyclin-dependent kinase-like 5 (CDKL-5) deficiency disorder.

In contrast to the relatively large number of antiseizure drugs that can attenuate seizure activity, there are currently no drugs known to prevent the formation of a seizure focus following CNS injury. Eventual development of such "antiepileptogenic" drugs will provide means of preventing the emergence of epilepsy following injuries such as head trauma, stroke, and CNS infection.

APPROACH TO THE PATIENT

Seizure

When a patient presents shortly after a seizure, the first priorities are attention to vital signs, respiratory and cardiovascular support, and treatment of seizures if they resume (see "Treatment: Seizures and Epilepsy"). Life-threatening conditions such as CNS infection, metabolic derangement, or drug toxicity must be recognized and managed appropriately.

When the patient is not acutely ill, the evaluation will initially focus on whether there is a history of earlier seizures **(Fig. 436-2)**. If this is the first seizure, then the emphasis will be to (1) establish whether the reported episode was a seizure rather than another paroxysmal event, (2) determine the cause of the seizure by identifying risk factors and precipitating events, and (3) decide whether antiseizure drug therapy is required in addition to treatment for any underlying illness.

In the patient with prior seizures or a known history of epilepsy, the evaluation is directed toward (1) identification of the underlying cause and precipitating factors, and (2) determination of the adequacy of the patient's current therapy.

HISTORY AND EXAMINATION

The first goal is to determine whether the event was truly a seizure. A detailed history is essential, because *in many cases the diagnosis of a seizure is based solely on clinical grounds—the examination and laboratory studies are often normal*. Questions should focus on the symptoms before, during, and after the episode to differentiate a seizure from other paroxysmal events (see "Differential Diagnosis of Seizures" below). Seizures frequently occur out-of-hospital, and the patient may be unaware of the ictal and immediate postictal phases; thus, witnesses to the event should be interviewed carefully.

The history should also focus on risk factors and predisposing events. Clues for a predisposition to seizures include a history of febrile seizures, a family history of seizures, and, of particular importance, earlier auras or brief seizures not recognized as such. Epileptogenic factors such as prior head trauma, stroke, tumor, or CNS infection should be identified. In children, a careful assessment of developmental

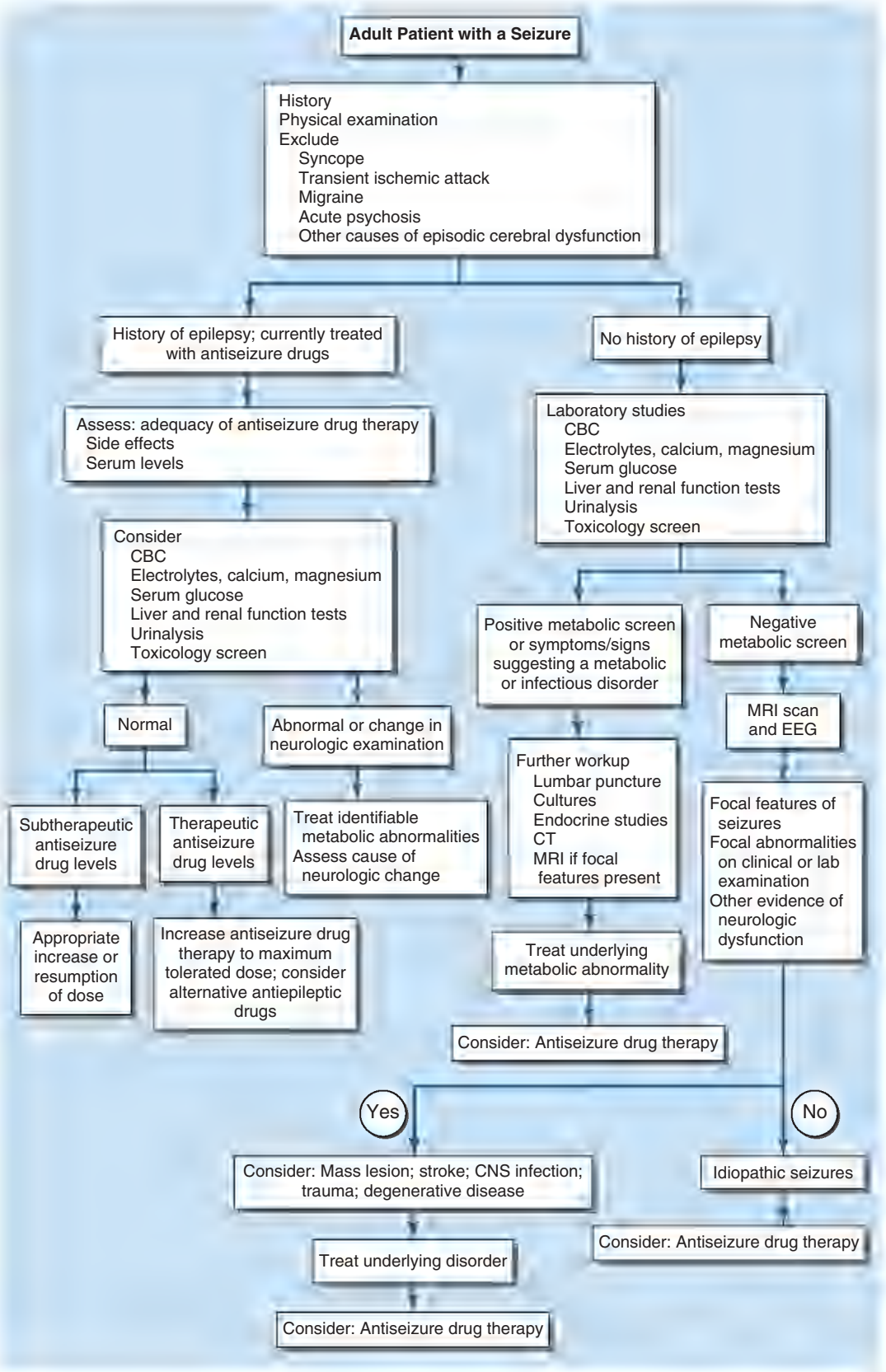

FIGURE 436-2 Evaluation of the adult patient with a seizure. CBC, complete blood count; CNS, central nervous system; CT, computed tomography; EEG, electroencephalogram; MRI, magnetic resonance imaging.

milestones may provide evidence for underlying CNS disease. Precipitating factors such as sleep deprivation, systemic diseases, electrolyte or metabolic derangements, acute infection, drugs that lower the seizure threshold (Table 436-5), or alcohol or illicit drug use should also be identified.

The general physical examination includes a search for signs of infection or systemic illness. Careful examination of the skin may reveal signs of neurocutaneous disorders, such as tuberous sclerosis or neurofibromatosis, or chronic liver or renal disease. A finding of organomegaly may indicate a metabolic storage disease, and limb asymmetry may provide a clue to brain injury early in development. Signs of head trauma and use of alcohol or illicit drugs should be sought. Auscultation of the heart and carotid arteries may identify an abnormality that predisposes to cerebrovascular disease.

All patients require a complete neurologic examination, with particular emphasis on eliciting signs of cerebral hemispheric disease (Chap. 433). Careful assessment of mental status (including memory, language function, and abstract thinking) may suggest lesions in the anterior frontal, parietal, or temporal lobes. Testing of visual fields will help screen for lesions in the optic pathways and occipital lobes. Screening tests of motor function such as pronator drift, deep tendon reflexes, gait, and coordination may suggest lesions in motor (frontal) cortex, and cortical sensory testing (e.g., double simultaneous stimulation) may detect lesions in the parietal cortex.

LABORATORY STUDIES

Routine blood studies are indicated to identify the more common metabolic causes of seizures such as abnormalities in electrolytes, glucose, calcium, or magnesium, and hepatic or renal disease. A screen for toxins in blood and urine should also be obtained from all patients in appropriate risk groups, especially when no clear precipitating factor has been identified. A lumbar puncture is indicated if there is any suspicion of meningitis or encephalitis, and it is mandatory in all patients infected with HIV, even in the absence of symptoms or signs suggesting infection. Testing for autoantibodies in the serum and cerebrospinal fluid (CSF) should be considered in patients presenting with fulminant onset of epilepsy associated with other abnormalities such as psychiatric symptoms or cognitive disturbances (Chap. 99).

ELECTROPHYSIOLOGIC STUDIES

The electrical activity of the brain (the EEG) is easily recorded from electrodes placed on the scalp. The potential difference between pairs of electrodes on the scalp (bipolar derivation) or between individual scalp electrodes and a relatively inactive common reference point (referential derivation) is amplified and displayed on a computer monitor. Digital systems allow the EEG to be displayed with any desired format and to be manipulated for more detailed analysis, and they also enable computational algorithms that can automatically detect certain abnormalities. The characteristics of the normal EEG depend on the patient's age and level of arousal. The rhythmic activity normally recorded represents the postsynaptic potentials of vertically oriented pyramidal cells of the cerebral cortex and is characterized by its frequency. In normal awake adults lying quietly with the eyes closed, an 8- to 13-Hz alpha rhythm is seen posteriorly in the EEG, intermixed with a variable amount of generalized faster (beta) activity (>13 Hz); the alpha rhythm is attenuated when the eyes are opened (Fig. 436-3). During drowsiness and light sleep, slower activity in the theta (4–7 Hz) and delta (<4 Hz) ranges becomes more conspicuous.

All patients who have a possible seizure disorder should be evaluated with an EEG as soon as possible. In the evaluation of a patient with suspected epilepsy, the presence of *electrographic seizure activity* during the clinical event (i.e., abnormal, repetitive, rhythmic activity having a discrete onset and termination) clearly establishes the diagnosis. The EEG is always abnormal during generalized tonic-clonic seizures. The absence of electrographic seizure activity does not exclude a seizure disorder, however, because focal seizures may originate from a region of the cortex that cannot be detected by standard scalp electrodes. Because seizures are typically infrequent and unpredictable, it is often not possible to obtain the EEG during a clinical event. In such situations, activating procedures are generally undertaken to provoke abnormalities while the EEG is recorded. These procedures commonly include hyperventilation, photic stimulation, sleep, and sleep deprivation on the night prior to the recording. Continuous monitoring for prolonged periods in video-EEG telemetry units for hospitalized patients or the use of portable equipment to record the EEG continuously for ≥24 h in ambulatory patients has made it easier to capture the electrophysiologic correlates of clinical events. Video-EEG telemetry is now a routine approach for the accurate diagnosis of epilepsy in patients with poorly characterized events or seizures that are difficult to control. A variety of technologies, including sub-scalp EEG devices, are being developed to enable ultra-long-term ambulatory recordings of brain activity, analogous to implanted loop recorders used to capture infrequent cardiac arrhythmias.

The EEG may also be helpful in the interictal period by showing certain abnormalities that are highly supportive of the diagnosis of epilepsy. Such *epileptiform activity* consists of bursts of abnormal discharges containing spikes or sharp waves. The presence of epileptiform activity is not entirely specific for epilepsy, but it has a much greater prevalence in patients with epilepsy than in other individuals. However, even in an individual who is known to have epilepsy, the initial routine interictal EEG may be normal 50–80% of the time. Thus, the EEG has limited sensitivity and cannot establish the diagnosis of epilepsy in many cases.

The EEG is also used for classifying seizure disorders and aiding in the selection of antiseizure medications (Fig. 436-4). For example, episodic generalized spike-wave activity is usually seen in patients with typical absence epilepsy and may be seen with other generalized epilepsy syndromes. Focal interictal epileptiform discharges would support the diagnosis of a focal seizure disorder such as temporal lobe epilepsy or frontal lobe seizures, depending on the location of the discharges.

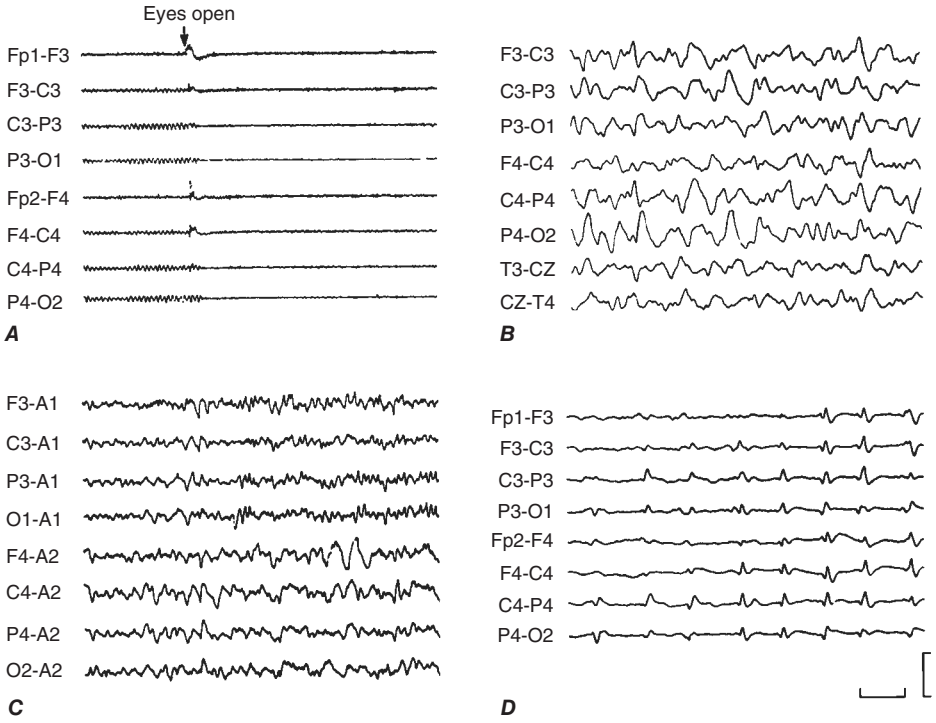

FIGURE 436-3 Electroencephalograms. ***A.*** Normal electroencephalogram (EEG) showing a posteriorly situated 9-Hz alpha rhythm that attenuates with eye opening. ***B.*** Abnormal EEG showing irregular diffuse slow activity in an obtunded patient with encephalitis. ***C.*** Irregular slow activity in the right central region, on a diffusely slowed background, in a patient with a right parietal glioma. ***D.*** Periodic complexes occurring once every second in a patient with Creutzfeldt-Jakob disease. Horizontal calibration: 1 s; vertical calibration: 200 μV in A, 300 μV in other panels. In this and the following figure, electrode placements are indicated at the left of each panel and accord with the international 10–20 system. A, earlobe; C, central; F, frontal; Fp, frontal polar; O, occipital; P, parietal; T, temporal. Right-sided placements are indicated by even numbers, left-sided placements by odd numbers, and midline placements by Z. *(Reproduced with permission from MJ Aminoff: Aminoff's Electrodiagnosis in Clinical Neurology, 6th ed. Oxford: Elsevier Saunders, 2012.)*

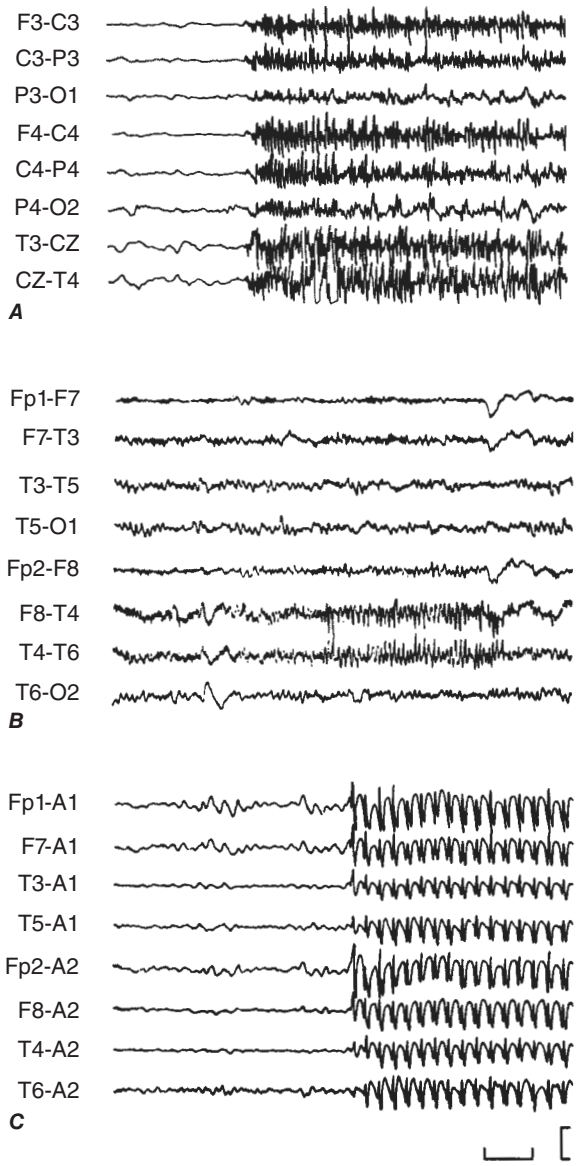

FIGURE 436-4 Electrographic seizures. ***A.*** Onset of a tonic seizure showing generalized repetitive sharp activity with synchronous onset over both hemispheres. ***B.*** Burst of repetitive spikes occurring with sudden onset in the right temporal region during a clinical spell characterized by transient impairment of awareness. ***C.*** Generalized 3-Hz spike-wave activity occurring synchronously over both hemispheres during an absence seizure. Horizontal calibration: 1 s; vertical calibration: 400 μV in *A*, 200 μV in *B*, and 750 μV in *C*. *(Reproduced with permission from MJ Aminoff: Aminoff's Electrodiagnosis in Clinical Neurology, 6th ed. Oxford: Elsevier Saunders, 2012.)*

The routine scalp EEG may also be used to assess the prognosis of seizure disorders; in general, a normal EEG implies a better prognosis, whereas an abnormal background or frequent epileptiform activity suggests a worse outcome. Unfortunately, the EEG has not proved to be useful in predicting which patients with predisposing conditions such as head injury or brain tumor will go on to develop epilepsy, because in such circumstances, epileptiform activity is commonly encountered regardless of whether seizures occur.

High-density EEG provides higher resolution localization by utilizing up to 257 scalp electrodes, as opposed to the 21 electrodes used in a standard EEG recording. This makes use of scalp voltage field data to determine dipole location, orientation, and propagation. EEG source imaging (ESI) permits visualization of these dipoles by mapping them onto a three-dimensional model that is co-registered on brain MRI.

Magnetoencephalography (MEG) provides another way of looking noninvasively at cortical activity. Instead of measuring electrical activity of the brain, it measures the small magnetic fields that are generated by this activity. The epileptiform activity seen on MEG can be analyzed, and its source in the brain can be estimated using a variety of mathematical techniques. These source estimates can then be plotted on an anatomic image of the brain such as an MRI (discussed below) to generate a magnetic source image (MSI). MSI can be useful to localize potential seizure foci.

BRAIN IMAGING

Almost all patients with new-onset seizures should have a brain imaging study to determine whether there is an underlying structural abnormality that is responsible. The only potential exception to this rule is children who have an unambiguous history and examination suggestive of a benign, generalized seizure disorder such as absence epilepsy. MRI has been shown to be superior to computed tomography (CT) for the detection of cerebral lesions associated with epilepsy. In some cases, MRI will identify lesions such as tumors, vascular malformations, or other pathologies that need urgent therapy. The availability of newer MRI methods, such as three-dimensional structural imaging at submillimeter resolution, has increased the sensitivity for detection of abnormalities of cortical architecture, including hippocampal atrophy associated with mesial temporal sclerosis, as well as abnormalities of neuronal migration. In such cases, the findings provide an explanation for the patient's seizures and point to the need for chronic antiseizure drug therapy or possible surgical resection.

In the patient with a suspected CNS infection or mass lesion, CT scanning should be performed emergently when MRI is not immediately available. Otherwise, it is usually appropriate to obtain an MRI study within a few days of the initial evaluation. Functional imaging procedures such as positron emission tomography (PET) and single-photon emission computed tomography (SPECT) are also used to evaluate certain patients with medically refractory seizures (Table 436-3).

GENETIC TESTING

With the increasing recognition of specific gene mutations causing epilepsy (Table 436-2), genetic testing is beginning to emerge as part of the diagnostic evaluation of patients with epilepsy. In addition to providing a definitive diagnosis (which may be of great benefit to the patient and family members and curtail the pursuit of additional, unrevealing laboratory testing), genetic testing may offer a guide for therapeutic options (see section "Selection of Antiseizure Drugs" below). Genetic testing is currently being done mainly in infants and children with epilepsy syndromes thought to have a genetic cause but should also be considered in older patients with a history suggesting an undiagnosed genetic epilepsy syndrome that began early in life.

DIFFERENTIAL DIAGNOSIS OF SEIZURES

Disorders that may mimic seizures are listed in **Table 436-6**. In most cases, seizures can be distinguished from other conditions by meticulous attention to the history and relevant laboratory studies. On occasion, additional studies such as video-EEG monitoring, sleep studies, tilt-table analysis, or cardiac electrophysiology may be required to reach a correct diagnosis. Two of the more common nonepileptic syndromes in the differential diagnosis are discussed below.

SYNCOPE

The diagnostic dilemma encountered most frequently is the distinction between a generalized seizure and syncope. Observations by the patient and bystanders that can help differentiate between the two are listed in **Table 436-7**. Characteristics of a seizure include the presence of an aura, cyanosis, unconsciousness, motor manifestations lasting >15 s, postictal disorientation, muscle soreness, and sleepiness. In contrast, a syncopal episode is more likely if the event was provoked by acute pain or emotional stress or occurred immediately after arising from the lying or sitting position. Patients with syncope often describe a stereotyped transition from consciousness to unconsciousness that includes tiredness, sweating, nausea, and tunneling of vision, and they experience a relatively brief loss of consciousness with rapid recovery of normal mentation. Headache and incontinence are unreliable

TABLE 436-6 Differential Diagnosis of Seizures

Syncope	**Transient ischemic attack (TIA)**
Vasovagal syncope	Basilar artery TIA
Cardiac arrhythmia	**Sleep disorders**
Valvular heart disease	Narcolepsy/cataplexy
Cardiac failure	Benign sleep myoclonus
Orthostatic hypotension	**Movement disorders**
Psychological disorders	Tics
Psychogenic seizure	Nonepileptic myoclonus
Hyperventilation	Paroxysmal choreoathetosis
Panic attack	**Special considerations in children**
Metabolic disturbances	Breath-holding spells
Alcoholic blackouts	Migraine with recurrent abdominal pain and cyclic vomiting
Delirium tremens	Benign paroxysmal vertigo
Hypoglycemia	Apnea
Hypoxia	Night terrors
Psychoactive drugs (e.g., hallucinogens)	Sleepwalking
Migraine	
Confusional migraine	
Basilar migraine	

features in differentiating between syncope and seizure. A brief period (i.e., 1–10 s) of convulsive motor activity is frequently seen immediately at the onset of a syncopal episode, especially if the patient remains in an upright posture after fainting (e.g., in a dentist's chair) and therefore has a sustained decrease in cerebral perfusion. Rarely, a syncopal episode can induce a full tonic-clonic seizure. In such cases, the evaluation must focus on both the cause of the syncopal event as well as the possibility that the patient has a propensity for recurrent seizures. Postictal symptoms can be helpful when differentiating convulsive syncope from seizure, as confusion and disorientation are typically much less prominent following syncope.

PSYCHOGENIC SEIZURES

Psychogenic seizures are nonepileptic behaviors that resemble seizures. They are often part of a conversion reaction precipitated by underlying psychological distress. Certain behaviors such as side-to-side turning of the head, ictal eye closure, asynchronous and large-amplitude shaking movements of the limbs, twitching of all four extremities without loss of consciousness, and pelvic thrusting are more commonly associated with psychogenic rather than epileptic seizures. Psychogenic seizures often last longer than epileptic seizures and may wax and wane over minutes to hours. However, the distinction is sometimes difficult on clinical grounds alone, and there are many examples of diagnostic errors made by experienced epileptologists. This is especially true for psychogenic seizures that resemble focal seizures, because the behavioral manifestations of focal seizures (especially of frontal lobe origin) can be extremely unusual, and, in both cases, the routine surface EEG may be normal. Video-EEG monitoring is very useful when historic features are nondiagnostic. Generalized tonic-clonic seizures always produce marked EEG abnormalities during and after the seizure. For suspected focal seizures, the use of additional electrodes may help to localize a seizure focus. Most generalized seizures and some focal seizures are accompanied by a rise in serum prolactin during the immediate 30-min postictal period, whereas psychogenic seizures are not, though this is not always reliable because baseline prolactin levels are rarely available and certain medications can elevate prolactin levels. The diagnosis of psychogenic seizures also does not exclude a concurrent diagnosis of epilepsy, because the two often coexist.

TABLE 436-7 Features That Distinguish Generalized Tonic-Clonic Seizure from Syncope

FEATURES	SEIZURE	SYNCOPE
Immediate precipitating factors	Usually none	Emotional stress, Valsalva, orthostatic hypotension, cardiac etiologies
Premonitory symptoms	None or aura (e.g., odd odor)	Tiredness, nausea, diaphoresis, tunneling of vision
Posture at onset	Variable	Usually erect
Transition to unconsciousness	Often immediate	Gradual over seconds[a]
Duration of unconsciousness	Minutes	Seconds
Duration of tonic or clonic movements	30–60 s	Never >15 s
Facial appearance during event	Cyanosis, frothing at mouth	Pallor
Disorientation and sleepiness after event	Many minutes to hours	<5 min
Aching of muscles after event	Often	Sometimes
Biting of tongue	Sometimes	Rarely
Incontinence	Sometimes	Sometimes
Headache	Sometimes	Rarely

[a]May be sudden with certain cardiac arrhythmias.

TREATMENT

Seizures and Epilepsy

Therapy for a patient with a seizure disorder is almost always multimodal and includes treatment of underlying conditions that cause or contribute to the seizures, avoidance of precipitating factors, suppression of recurrent seizures by prophylactic therapy with antiseizure medications or surgery, and addressing a variety of psychological and social issues. Treatment plans must be individualized, given the many different types and causes of seizures as well as the differences in efficacy and toxicity of antiseizure medications for each patient. In almost all cases, a neurologist with experience in the treatment of epilepsy should design and oversee implementation of the treatment strategy. Furthermore, patients with refractory epilepsy or those who require polypharmacy with antiseizure drugs should remain under the regular care of a neurologist.

TREATMENT OF UNDERLYING CONDITIONS

If the sole cause of a seizure is a metabolic disturbance such as an abnormality of serum electrolytes or glucose, then treatment is aimed at reversing the metabolic problem and preventing its recurrence. Therapy with antiseizure drugs is usually unnecessary unless the metabolic disorder cannot be corrected promptly and the patient is at risk of having further seizures. If the apparent cause of a seizure was a medication (e.g., bupropion) or illicit drug use (e.g., cocaine), then appropriate therapy is avoidance of the drug; there is usually no need for antiseizure medications unless subsequent seizures occur in the absence of these precipitants.

Seizures caused by a structural CNS lesion such as a brain tumor, vascular malformation, or brain abscess may not recur after appropriate treatment of the underlying lesion. However, despite removal of the structural lesion, there is a risk that the seizure focus will remain in the surrounding tissue or develop de novo as a result of gliosis and other processes induced by surgery, radiation, or other therapies. Most patients are therefore maintained on an antiseizure medication for 6–12 months, and an attempt is made to withdraw medications only if the patient has been completely seizure free. If seizures are refractory to medication, the patient may benefit from surgical removal of the seizure-producing brain tissue (see below).

AVOIDANCE OF PRECIPITATING FACTORS

Unfortunately, little is known about the specific factors that determine precisely when a seizure will occur in a patient with epilepsy. An almost universal precipitating factor for seizures is sleep deprivation, so patients should do everything possible to optimize their sleep quality. Many patients can identify other avoidable situations that appear to lower their seizure threshold. For example, patients may note an association between alcohol intake and seizures, and

they should be encouraged to modify their drinking habits accordingly. There are also relatively rare cases of patients with seizures that are induced by highly specific stimuli such as a video game monitor, music, startling sounds, or an individual's voice ("reflex epilepsy"). Because there is often an association between stress and seizures, stress reduction techniques such as physical exercise, meditation, or counseling may be helpful.

ANTISEIZURE DRUG THERAPY

Antiseizure drug therapy is the mainstay of treatment for most people with epilepsy. The overall goal is to completely prevent seizures without causing any untoward side effects, preferably with a single medication and a dosing schedule that is easy for the patient to follow. Seizure classification is an important element in designing the treatment plan, because some antiseizure drugs have different activities against various seizure types. However, there is considerable overlap between antiseizure drugs, and choice of therapy is often determined by anticipated side effects, drug-drug interactions, medical comorbidities, dosing frequency, and cost.

When to Initiate Antiseizure Drug Therapy Antiseizure drug therapy should be started in any patient with recurrent seizures of unknown etiology or a known cause that cannot be reversed. Whether to initiate therapy in a patient with a single seizure is controversial. Patients with a single seizure due to an identified lesion such as a CNS tumor, infection, or trauma, in which there is strong evidence that the lesion is epileptogenic, should be treated. The risk of seizure recurrence in a patient with an apparently unprovoked seizure is uncertain, with estimates ranging from 31 to 71% in the first 12 months after the initial seizure. This uncertainty arises from differences in the underlying seizure types and etiologies in various published epidemiologic studies. Generally accepted risk factors associated with recurrent seizures include the following: (1) prior brain insult such as a stroke or trauma, (2) an EEG with epileptiform abnormalities, (3) a significant brain imaging abnormality, or (4) a nocturnal seizure. Most patients with one or more of these risk factors should be treated. Issues such as employment or driving may influence the decision regarding whether to start medications as well. For example, a patient with a single unprovoked seizure whose job depends on driving may prefer taking an antiseizure drug to reduce risk of seizure recurrence and the potential loss of driving privileges.

Selection of Antiseizure Drugs Antiseizure drugs available in the United States are shown in **Table 436-8**, and the main pharmacologic characteristics of commonly used drugs are listed in **Table 436-9**. Worldwide, older medications such as phenytoin, valproic acid, carbamazepine, phenobarbital, and ethosuximide are generally used as first-line therapy for most seizure disorders because, overall, they are as effective as more recent drugs and significantly less expensive overall. Most of the new drugs that have become available in the past decade are used as adjunctive therapy, although many are now also being used as first-line monotherapy.

In addition to efficacy, factors influencing the choice of an initial medication include the convenience of dosing (e.g., once daily vs three times daily) and potential side effects. In this regard, many of the newer drugs have the advantage of reduced drug-drug interactions and easier dosing. Almost all the commonly used antiseizure drugs can cause similar, dose-related side effects such as sedation, ataxia, dizziness, and diplopia. Long-term use of some agents in adults, especially the elderly, can lead to osteoporosis. Close follow-up is required to ensure these side effects are promptly recognized and reversed. Most of the older drugs and some of the newer ones can also cause idiosyncratic toxicity such as rash, bone marrow suppression, or hepatotoxicity. Although rare, these side effects should be considered during drug selection, and patients must be instructed about symptoms or signs that should signal the need to alert their health care provider. For some drugs, laboratory tests (e.g., complete blood count and liver function tests) are recommended prior to the institution of therapy (to establish baseline values) and during initial dosing and titration of the agent. Monitoring serum concentrations of antiseizure medications can help determine when a therapeutic dose has been reached, though clinical response is paramount (see below).

TABLE 436-8 Selection of Antiseizure Drugs

GENERALIZED-ONSET TONIC-CLONIC	FOCAL	TYPICAL ABSENCE	ATYPICAL ABSENCE, MYOCLONIC, ATONIC
First-Line			
Lamotrigine	Lamotrigine	Valproic acid	Valproic acid
Valproic acid	Carbamazepine	Ethosuximide	Lamotrigine
	Oxcarbazepine	Lamotrigine	Topiramate
	Eslicarbazepine		
	Phenytoin		
	Levetiracetam		
Alternatives			
Zonisamide[a]	Zonisamide[a]	Clonazepam	Clonazepam
Phenytoin	Brivaracetam	Zonisamide	Felbamate
Levetiracetam	Topiramate	Levetiracetam	Clobazam
Carbamazepine	Valproic acid		Rufinamide
Oxcarbazepine	Tiagabine[a]		Fenfluramine
Topiramate	Gabapentin[a]		
Phenobarbital	Lacosamide[a]		
Primidone	Phenobarbital		
Felbamate	Primidone		
Perampanel	Felbamate		
	Perampanel		
	Cenobamate[a]		

[a]As adjunctive therapy.

An important advance in the care of people with epilepsy has been the application of genetic testing to help guide the choice of therapy (as well as establishing the underlying cause of a patient's syndrome; Table 436-2). For example, the identification of a mutation in the *SLC2A1* gene, which encodes the glucose type 1 transporter (GLUT-1) and is a cause of GLUT-1 deficiency, should prompt immediate treatment with the ketogenic diet. Mutations of the *ALDH7A1* gene, which encodes antiquitin, can cause alterations in pyridoxine metabolism that are reversed by treatment with pyridoxine. There is also mounting evidence that certain gene mutations may indicate better or worse response to specific antiseizure drugs. For example, patients with mutations in the sodium channel subunit *SCN1A* should generally avoid taking phenytoin or lamotrigine, whereas patients with mutations in the *SCN2A* or *SCN8A* sodium channel subunits appear to respond favorably to high-dose phenytoin. Genetic testing may also help predict antiseizure drug toxicity. Studies have shown that individuals of Asian descent who carry the human leukocyte antigen (HLA) allele HLA-B*1502 are at particularly high risk of developing serious skin reactions from carbamazepine, phenytoin, oxcarbazepine, and lamotrigine. HLA-A*31:01 has also been found to be associated with carbamazepine-induced hypersensitivity reactions in patients of European or Japanese ancestry.

Antiseizure Drug Selection for Focal Seizures Carbamazepine (or related drugs, oxcarbazepine and eslicarbazepine), lamotrigine, phenytoin, and levetiracetam are currently the drugs of choice approved for the initial treatment of focal seizures, including those that evolve into generalized seizures. Overall, they have very similar efficacy, but differences in pharmacokinetics and toxicity are the main determinants for use in a given patient. For example, an advantage of carbamazepine (which is also available in an extended-release form) is that its metabolism follows first-order pharmacokinetics, which allows for a linear relationship between drug dose, serum levels, and toxicity. Carbamazepine can cause leukopenia, aplastic anemia, or hepatotoxicity and would therefore be

TABLE 436-9 Dosage and Adverse Effects of Commonly Used Antiepileptic Drugs

GENERIC NAME	TRADE NAME	PRINCIPAL USES	TYPICAL DOSE; DOSE INTERVAL	HALF-LIFE	THERAPEUTIC RANGE	ADVERSE EFFECTS		DRUG INTERACTIONS[a]
						NEUROLOGIC	SYSTEMIC	
Brivaracetam	Briviact	Focal onset	100–200 mg/d; bid	7–10 h	Not established	Fatigue Dizziness Weakness Ataxia Mood changes	Gastrointestinal irritation	May increase carbamazepine-epoxide causing decreased tolerability May increase phenytoin
Cannabidiol	Epidiolex	Dravet and Lennox-Gastaut syndromes Tuberous sclerosis complex-associated seizures	10–20 mg/kg per d; bid	18–32 h	Not established	Sedation	Elevated transaminases Anorexia Weight loss Diarrhea	Increases clobazam causing somnolence
Carbamazepine	Tegretol[c]	Tonic-clonic Focal onset	600–1800 mg/d (15–35 mg/kg, child); bid (capsules or tablets), tid-qid (oral suspension)	10–17 h (variable due to autoinduction: complete 3–5 wk after initiation)	4–12 μg/mL	Ataxia Dizziness Diplopia Vertigo	Aplastic anemia Leukopenia Gastrointestinal irritation Hepatotoxicity Hyponatremia Rash	Level decreased by enzyme-inducing drugs[b] Level increased by erythromycin, propoxyphene, isoniazid, cimetidine, fluoxetine
Cenobamate	Xcopri	Focal onset	100–400 mg/d; daily (tablets)	50–60 h	Not established	Cognitive dysfunction Dizziness Disequilibrium Gait disturbance Headache	Anorexia Constipation Diarrhea Dyspepsia Nausea	Major CYP3A4 inducer; moderate CYP2C19 inhibitor
Clobazam	Onfi	Lennox-Gastaut syndrome	10–40 mg/d (5–20 mg/d for patients <30 kg body weight); bid	36–42 h (71–82 h for less active metabolite)	Not established	Fatigue Sedation Ataxia Aggression Insomnia	Constipation Anorexia Skin rash	Level increased by CYP2C19 inhibitors
Clonazepam	Klonopin	Absence Atypical absence Myoclonic	1–12 mg/d; qd-tid	24–48 h	10–70 ng/mL	Ataxia Sedation Lethargy	Anorexia	Level decreased by enzyme-inducing drugs[b]
Eslicarbazepine	Aptiom	Focal onset	400–1600 mg/d; qd	20–24 h	10–35 μg/mL (as oxcarbazepine mono-hydroxy derivative)	Sedation Ataxia Dizziness Diplopia Vertigo	See carbamazepine	Level decreased by enzyme-inducing drugs[b]
Ethosuximide	Zarontin	Absence	750–1250 mg/d (20–40 mg/kg); qd-bid	60 h, adult 30 h, child	40–100 μg/mL	Ataxia Lethargy Headache	Gastrointestinal irritation Skin rash Bone marrow suppression	Level decreased by enzyme-inducing drugs[b] Level increased by valproic acid
Felbamate	Felbatol	Focal onset Lennox-Gastaut syndrome Tonic-clonic	2400–3600 mg/d, tid-qid	16–22 h	30–60 μg/mL	Insomnia Dizziness Sedation Headache	Aplastic anemia Hepatic failure Weight loss Gastrointestinal irritation	Increases phenytoin, valproic acid, active carbamazepine metabolite
Fenfluramine	Fintepla	Dravet and Lennox-Gastaut syndromes	0.1–0.35 mg/kg/dose bid (oral solution); dosage depends on coadministration with stiripentol and/or clobazam	20 h	Not established	Ataxia Behavioral disturbance Headache Somnolence	Anorexia Constipation Hypertension Serotonin syndrome Weight loss	CYP1A2, CYP2B6, CYP2D6, CYPDA4 substrate

(*Continued*)

TABLE 436-9 Dosage and Adverse Effects of Commonly Used Antiepileptic Drugs (Continued)

GENERIC NAME	TRADE NAME	PRINCIPAL USES	TYPICAL DOSE; DOSE INTERVAL	HALF-LIFE	THERAPEUTIC RANGE	ADVERSE EFFECTS		DRUG INTERACTIONS[a]
						NEUROLOGIC	SYSTEMIC	
Gabapentin	Neurontin	Focal onset	900–2400 mg/d; tid-qid	5–9 h	2–20 μg/mL	Sedation Dizziness Ataxia Fatigue	Gastrointestinal irritation Weight gain Edema	No known significant interactions
Ganaxolone	Ztalmy	CDKL-5 deficiency disorder–associated seizures	450–1800 mg/d; tid (oral suspension)	34 h	Not established	Gait disturbance Somnolence	Bronchitis Fever Nasal congestion	CYP2B6, CYP2C19, CYP3A4 substrate
Lacosamide	Vimpat	Focal onset	200–400 mg/d; bid	13 h	Not established	Dizziness Ataxia Diplopia Vertigo	Gastrointestinal irritation Cardiac conduction (PR interval prolongation)	Level decreased by enzyme-inducing drugs[b]
Lamotrigine	Lamictal[c]	Focal onset Tonic-clonic Atypical absence Myoclonic Lennox-Gastaut syndrome	150–500 mg/d; bid (immediate release), daily (extended release) (lower daily dose for regimens with valproic acid; higher daily dose for regimens with an enzyme inducer)	25 h 14 h (with enzyme inducers), 59 h (with valproic acid)	2.5–20 μg/mL	Dizziness Diplopia Sedation Ataxia Headache	Skin rash Stevens-Johnson syndrome	Level decreased by enzyme-inducing drugs[b] and oral contraceptives Level increased by valproic acid
Levetiracetam	Keppra	Focal onset	1000–3000 mg/d; bid (immediate release), daily (extended release)	6–8 h	5–45 μg/mL	Sedation Fatigue Incoordination Mood changes	Anemia Leukopenia	No known significant interactions
Oxcarbazepine[c]	Trileptal	Focal onset Tonic-clonic	900–2400 mg/d (30–45 mg/kg, child); bid	10–17 h (for active metabolite)	10–35 μg/mL	Fatigue Ataxia Dizziness Diplopia Vertigo Headache	See carbamazepine	Level decreased by enzyme-inducing drugs[b] May increase phenytoin
Perampanel	Fycompa	Focal onset Tonic-clonic	4–12 mg; qd	105 h	Not established	Dizziness Somnolence Aggression Ataxia Anxiety Paranoia	Headache Nausea	Level decreased by enzyme-inducing drugs[b]
Phenobarbital	Luminal	Tonic-clonic Focal onset	60–180 mg/d; qd-tid	90 h	10–40 μg/mL	Sedation Ataxia Confusion Dizziness Decreased libido Depression	Skin rash	Level increased by valproic acid, phenytoin
Phenytoin[c] (diphenylhydantoin)	Dilantin	Tonic-clonic Focal onset	300–400 mg/d (3–6 mg/kg, adult; 4–8 mg/kg, child); qd-tid	24 h (wide variation, dose-dependent)	10–20 μg/mL	Dizziness Diplopia Ataxia Incoordination Confusion	Gingival hyperplasia Lymphadenopathy Hirsutism Osteomalacia Facial coarsening Skin rash	Level increased by isoniazid, sulfonamides, fluoxetine Level decreased by enzyme-inducing drugs[b] Altered folate metabolism
Primidone	Mysoline	Tonic-clonic Focal onset	750–1000 mg/d; bid-tid	Primidone, 8–15 h Phenobarbital, 90 h	Primidone, 4–12 μg/mL Phenobarbital, 10–40 μg/mL	Same as phenobarbital		Level increased by valproic acid Level decreased by phenytoin (increased conversion to phenobarbital)

(*Continued*)

TABLE 436-9 Dosage and Adverse Effects of Commonly Used Antiepileptic Drugs (Continued)

GENERIC NAME	TRADE NAME	PRINCIPAL USES	TYPICAL DOSE; DOSE INTERVAL	HALF-LIFE	THERAPEUTIC RANGE	ADVERSE EFFECTS		DRUG INTERACTIONS[a]
						NEUROLOGIC	SYSTEMIC	
Rufinamide	Banzel	Lennox-Gastaut syndrome	3200 mg/d (45 mg/kg, child); bid	6–10 h	Not established	Sedation Fatigue Dizziness Ataxia Headache Diplopia	Gastrointestinal irritation Leukopenia Cardiac conduction (QT interval shortening)	Level decreased by enzyme-inducing drugs[b] Level increased by valproic acid May increase phenytoin
Tiagabine	Gabitril	Focal onset	32–56 mg/d; bid-qid (as adjunct to enzyme-inducing antiepileptic drug regimen)	2–5 h (with enzyme inducer), 7–9 h (without enzyme inducer)	Not established	Confusion Sedation Depression Dizziness Speech or language problems Paresthesias Psychosis	Gastrointestinal irritation	Level decreased by enzyme-inducing drugs[b]
Topiramate[c]	Topamax	Focal onset Tonic-clonic Lennox-Gastaut syndrome	200–400 mg/d; bid (immediate release), daily (extended release)	20 h (immediate release), 30 h (extended release)	2–20 μg/mL	Psychomotor slowing Sedation Speech or language problems Fatigue Paresthesias	Renal stones (avoid use with other carbonic anhydrase inhibitors) Glaucoma Weight loss Hypohidrosis	Level decreased by enzyme-inducing drugs[b]
Valproic acid (valproate sodium, divalproex sodium)	Depakene Depakote	Tonic-clonic Absence Atypical absence Myoclonic Focal onset Atonic	750–2000 mg/d (20–60 mg/kg); bid-qid (immediate and delayed release), daily (extended release)	15 h	50–125 μg/mL	Ataxia Sedation Tremor	Hepatotoxicity Thrombocytopenia Gastrointestinal irritation Weight gain Transient alopecia Hyperammonemia	Level decreased by enzyme-inducing drugs[b]
Zonisamide	Zonegran	Focal onset Tonic-clonic	200–400 mg/d; qd-bid	50–68 h	10–40 μg/mL	Sedation Dizziness Confusion Headache Psychosis	Anorexia Renal stones Hypohidrosis	Level decreased by enzyme-inducing drugs[b]

[a]Examples only; please refer to other sources for comprehensive listings of all potential drug-drug interactions. [b]Phenytoin, carbamazepine, phenobarbital. [c]Extended-release product available.

contraindicated in patients with predispositions to these problems. Oxcarbazepine has the advantage of being metabolized in a way that avoids an intermediate metabolite ("toxic epoxide") associated with some of the side effects of carbamazepine. Oxcarbazepine also has fewer drug interactions than carbamazepine. Eslicarbazepine has a long serum half-life and is dosed once daily.

Lamotrigine tends to be well tolerated in terms of side effects and has mood-stabilizing properties that can be beneficial. However, patients need to be particularly vigilant about the possibility of a skin rash during the initiation of therapy. This can be extremely severe and lead to Stevens-Johnson syndrome if unrecognized and if the medication is not discontinued immediately. This risk can be reduced by using low initial doses and slow titration. Lamotrigine must be started at even lower initial doses when used as add-on therapy with valproic acid, because valproic acid inhibits lamotrigine metabolism and substantially prolongs its half-life.

Phenytoin has a relatively long half-life and offers the advantage of once- or twice-daily dosing compared to twice- or thrice-daily dosing for many of the other drugs. However, phenytoin shows properties of nonlinear kinetics, such that small increases in phenytoin doses above a standard maintenance dose can precipitate marked side effects. This is one of the main causes of acute phenytoin toxicity (dizziness, diplopia, ataxia). Long-term use of phenytoin is associated with untoward cosmetic effects (e.g., hirsutism, coarsening of facial features, gingival hypertrophy) and osteoporosis. Due to these side effects, phenytoin is often avoided in young patients who are likely to require the drug for many years.

Levetiracetam has the advantage of having no known clinically relevant drug-drug interactions, making it especially useful in the elderly and patients on other medications. However, a significant number of patients taking levetiracetam complain of irritability, anxiety, and other psychiatric symptoms.

Topiramate can be used for both focal and generalized seizures. Like some of the other antiseizure drugs, topiramate can cause significant psychomotor slowing and other cognitive problems. Additionally, it should not be used in patients at risk for renal stones.

Valproic acid is an effective alternative for some patients with focal seizures, especially when the seizures generalize. Gastrointestinal side effects are fewer when using the delayed-release formulation. Laboratory testing is required to monitor toxicity because valproic acid can rarely cause reversible bone marrow suppression and hepatotoxicity. This drug should generally be avoided in patients with preexisting bone marrow or liver disease. Valproic acid also has relatively high risks of unacceptable adverse effects for women of childbearing age, including hyperandrogenism, that may affect fertility and teratogenesis (e.g., neural tube defects) in offspring. Irreversible, fatal hepatic failure appearing as an idiosyncratic rather than dose-related side effect is a relatively

rare complication; its risk is highest in children <2 years old, especially those taking other antiseizure drugs or with inborn errors of metabolism.

Zonisamide, brivaracetam, tiagabine, gabapentin, perampanel, and lacosamide are additional drugs currently used for the treatment of focal seizures with or without evolution into generalized seizures. Phenobarbital and other barbiturate compounds were commonly used in the past as first-line therapy for many forms of epilepsy. However, the barbiturates frequently cause sedation in adults, hyperactivity in children, and other more subtle cognitive changes; thus, their use should be limited to situations in which no other suitable treatment alternatives exist. Cenobamate is a recently approved antiseizure drug that has been shown to significantly improve seizure control in patients with focal epilepsy who were not adequately treated with up to three medications.

Antiseizure Drug Selection for Generalized Seizures Lamotrigine, valproic acid, and levetiracetam are currently considered the best initial choice for the treatment of primary generalized, tonic-clonic seizures. Topiramate, zonisamide, perampanel, phenytoin, carbamazepine, and oxcarbazepine are suitable alternatives, although carbamazepine, oxcarbazepine, and phenytoin can worsen certain types of generalized seizures. Valproic acid is particularly effective in absence, myoclonic, and atonic seizures. It is therefore commonly used in patients with generalized epilepsy syndromes having mixed seizure types. However, levetiracetam, rather than valproic acid, is increasingly considered the initial drug of choice for women with epilepsies having mixed seizure types given the adverse effects of valproic acid for women of childbearing age (discussed below). Lamotrigine is also an alternative to valproate, especially for absence epilepsies. Ethosuximide is a particularly effective drug for the treatment of absence seizures, but it is not useful for tonic-clonic or focal seizures. Periodic monitoring of blood cell counts is required since ethosuximide rarely causes bone marrow suppression.

INITIATION AND MONITORING OF THERAPY

Because the response to any antiseizure drug is unpredictable, patients should be carefully educated about the approach to therapy. The goal is to prevent seizures and minimize the side effects of treatment; determination of the optimal medication and the optimal dose typically involves trial and error. This process may take months or longer if the baseline seizure frequency is low. Most antiseizure drugs need to be introduced relatively slowly to minimize side effects. Patients should expect that minor side effects such as mild sedation, slight changes in cognition, or imbalance will typically resolve within a few days. Starting doses are usually the lowest value listed under the dosage column in Table 436-9. Subsequent increases should be made only after achieving a steady state with the previous dose (i.e., after an interval of five or more half-lives).

Monitoring of serum antiseizure drug levels can be very useful for establishing the initial dosing schedule. However, the published therapeutic ranges of serum drug concentrations are only an approximate guide for determining the proper dose for a given patient. The key determinants are the clinical measures of seizure frequency and presence of side effects, not the laboratory values. Conventional assays of serum drug levels measure the total drug (i.e., both free and protein bound). However, it is the concentration of free drug that reflects extracellular levels in the brain and correlates best with efficacy. Thus, patients with decreased levels of serum proteins (e.g., decreased serum albumin due to impaired liver or renal function) may have an increased ratio of free to bound drug, yet the concentration of free drug may be adequate for seizure control. These patients may have a "subtherapeutic" drug level, but the dose should be changed only if seizures remain uncontrolled, not just to achieve a "therapeutic" level. It is also useful to monitor free drug levels in such patients. In practice, other than during the initiation or modification of therapy, monitoring of antiseizure drug levels is most useful for documenting adherence, assessing clinical suspicion of toxicity, or establishing baseline serum concentrations prior to pregnancy, when clearance of many antiseizure drugs increases significantly.

If seizures continue despite gradual increases to the maximum tolerated dose and documented compliance, then it becomes necessary to switch to another antiseizure drug. This is usually done by maintaining the patient on the first drug while a second drug is added. The dose of the second drug should be adjusted to decrease seizure frequency without causing toxicity. Once this is achieved, the first drug can be gradually withdrawn (usually over weeks unless there is significant toxicity). The dose of the second drug is then further optimized based on seizure response and side effects. Monotherapy should be the goal whenever possible.

WHEN TO DISCONTINUE THERAPY

Some patients who have their seizures completely controlled with antiseizure drugs can eventually discontinue therapy. The following patient profile yields the greatest chance of remaining seizure free after drug withdrawal: (1) complete medical control of seizures for 1–5 years; (2) single seizure type, with generalized seizures having a better prognosis than focal seizures; (3) normal neurologic examination, including intelligence; (4) no family history of epilepsy; and (5) normal EEG. The appropriate seizure-free interval is unknown and depends on the form of epilepsy and whether or not the causal factor is still present (e.g., resection of a brain tumor causing seizures). However, it seems reasonable to attempt withdrawal of therapy after 2 years in a patient who meets all the above criteria, is motivated to discontinue the medication, and clearly understands the potential risks and benefits. In most cases, it is preferable to reduce the dose of the drug gradually over 2–3 months. Most recurrences occur in the first 3 months after discontinuing therapy, and patients should be advised to avoid potentially dangerous situations such as driving or swimming during this period. Rarely, seizure-free patients who discontinue antiseizure medications but then have a recurrent seizure may not regain full control when these medications are resumed.

TREATMENT OF REFRACTORY EPILEPSY

Approximately half of people with epilepsy do not respond to treatment with the first antiseizure drug, and it becomes necessary to try additional drugs alone or in combination. Patients who have focal epilepsy related to an underlying structural lesion or those with multiple seizure types and developmental delay are particularly likely to require multiple drugs. There are currently no clear guidelines for rational polypharmacy, although in theory, a combination of drugs with different mechanisms of action may be most useful. In most cases, the initial combination therapy combines first-line drugs (i.e., carbamazepine, oxcarbazepine, lamotrigine, valproic acid, levetiracetam, and phenytoin). If these drugs are unsuccessful, then the addition of other drugs such as cenobamate, zonisamide, brivaracetam, topiramate, or lacosamide is indicated. Patients with myoclonic seizures resistant to valproic acid may benefit from the addition of levetiracetam, zonisamide, clonazepam, or clobazam, and those with absence seizures may respond to a combination of valproic acid and ethosuximide. The same principles concerning the monitoring of therapeutic response, toxicity, and serum levels for monotherapy apply to polypharmacy, and potential drug interactions need to be recognized. If there is no improvement, a third drug can be added while the first two are maintained. If there is a response, the less effective or less well tolerated of the first two drugs should be gradually withdrawn.

SURGICAL TREATMENT OF REFRACTORY EPILEPSY

Approximately 30% of patients with epilepsy continue to have seizures despite efforts to find an effective combination of antiseizure drugs. For some patients with focal epilepsy, surgery can be extremely effective in substantially reducing seizure frequency and even providing complete seizure control. Understanding the potential value of surgery is especially important when a patient's seizures are not controlled with initial treatment, as such patients

often do not respond to subsequent medication trials. Rather than submitting the patient to years of unsuccessful medical therapy and the psychosocial trauma and increased mortality associated with ongoing seizures, the patient should have an efficient but relatively brief attempt at medical therapy and then be referred for surgical evaluation.

The most common surgical procedure for patients with temporal lobe epilepsy involves resection of the anteromedial temporal lobe (temporal lobectomy) or a more limited removal of the underlying hippocampus and amygdala (amygdalohippocampectomy). Focal seizures arising from extratemporal regions may be abolished by a focal neocortical resection with precise removal of an identified lesion (lesionectomy). Localized neocortical resection without a clear lesion identified on MRI is also possible when other tests (e.g., MEG, PET, SPECT) implicate a focal cortical region as a seizure onset zone. When the cortical region cannot be removed, multiple subpial transection, which disrupts intracortical connections, is sometimes used to prevent seizure spread. Hemispherectomy or multilobar resection is useful for some patients with severe seizures due to hemispheric abnormalities such as hemimegalencephaly or other dysplastic abnormalities, and corpus callosotomy has been shown to be effective for disabling tonic or atonic seizures, usually when they are part of a mixed-seizure syndrome (e.g., Lennox-Gastaut syndrome).

Presurgical evaluation is designed to identify the functional and structural basis of the patient's seizure disorder. Inpatient video-EEG monitoring is used to localize the seizure focus and to correlate the abnormal electrophysiologic activity with neuroimaging and behavioral manifestations of the seizure. Routine scalp EEG recordings and a high-resolution MRI scan are often sufficient for localization of the epileptogenic focus, especially when the findings are concordant. Functional imaging studies such as SPECT, PET, and MEG are adjunctive tests that may help to reveal or verify the localization of an apparent epileptogenic region. Once the presumed location of the seizure onset is identified, additional studies, including neuropsychological testing, the intracarotid amobarbital (Wada) test, and functional MRI may be used to assess language and memory localization and to determine the possible functional consequences of surgical removal of the epileptogenic region. In some cases, standard noninvasive evaluation is not sufficient to localize the seizure onset zone, and invasive electrophysiologic monitoring is required for more definitive localization. Traditionally, this required open craniotomy and subdural electrode placement. Robot-assisted stereotactic EEG (stereo-EEG) has surged in use as a less invasive surgical option that involves placing depth electrodes through burr holes in the skull and into the brain parenchyma to record from regions suspected of generating seizures. Although subdural electrodes provide extensive cortical sampling, they are unable to record from deep structures; depth electrodes permit sampling of both cortical and subcortical tissue, albeit with more restricted spatial resolution. Stereo-EEG studies are typically associated with shorter hospital stays and lower complication rates than subdural electrode studies.

The exact extent of the resection to be undertaken can also be determined by performing cortical mapping at the time of the surgical procedure, allowing for a tailored resection. This involves electrocorticographic recordings made with electrodes on the surface of the brain to identify the extent of epileptiform disturbances. If the region to be resected is within or near brain regions suspected of having sensorimotor or language function, electrical cortical stimulation mapping is performed on the awake patient to determine the function of cortical regions in question to avoid resection of so-called eloquent cortex and thereby minimize postsurgical deficits.

Advances in presurgical evaluation and microsurgical techniques have led to a steady increase in the success of epilepsy surgery. Clinically significant complications of surgery are <5%, and the use of functional mapping procedures has markedly reduced the neurologic sequelae due to removal or sectioning of brain tissue. For example, ~70% of well-selected patients treated with temporal lobectomy will become seizure free, and another 15–25% will have at least a 90% reduction in seizure frequency. Marked improvement is also usually seen in patients treated with hemispherectomy for catastrophic seizure disorders due to large hemispheric abnormalities. Postoperatively, patients generally need to remain on antiseizure drug therapy, but the marked reduction of seizures following resective surgery can have a very beneficial effect on quality of life. Recently, catheter-based stereotactic laser thermal ablation has been developed as a less invasive means for destroying the seizure focus in select patients.

Not all medically refractory patients are suitable candidates for resective surgery or laser ablation. For example, some patients have seizures arising from more than one brain region or from a single "eloquent" region that mediates a critical function (e.g., vision, movement, language), such that the potential harm from removal is unacceptably high. In these patients, implanted neurostimulation devices that deliver electrical energy to the brain to reduce seizures represent palliative treatment options. Vagus nerve stimulation (VNS) involves an extracranial device that works through scheduled intermittent ("open loop") stimulation of the left vagus nerve. Efficacy of VNS is limited, and side effects related to recurrent laryngeal nerve activation (e.g., hoarseness, throat pain, dyspnea) can be significant and dose-limiting. By contrast, responsive neurostimulation (RNS) involves an implanted device connected to two lead wires that are placed intracranially at the site(s) from where seizures arise. The neurostimulator detects the onset of a seizure (often before the seizure becomes clinically apparent) and delivers electrical stimulation—typically imperceptible—directly to the brain to reduce seizures over time, a form of "closed loop" neurostimulation. RNS is the only device that provides chronic EEG, which has a growing number of clinical applications, such as quantifying the lateralization of seizures arising from both sides of the brain, characterizing clinical spells, assessing effects of medications and other therapeutic interventions, and revealing cyclical patterns of epileptic brain activity that may help anticipate future events. A third modality, thalamic deep brain stimulation (DBS), involves open loop stimulation of deep, bilateral cerebral structures, the anterior thalamic nuclei, which are key nodes in limbic circuits mediating certain types of seizures. Whereas precise seizure localization is necessary for RNS, it is not required for VNS or DBS. Stimulation of thalamic nuclei, which project broadly to different cortical regions, with RNS or DBS is an appealing approach to treating poorly localized, spatially extensive, or multifocal seizure foci.

Long-term clinical trials of all three neurostimulation devices demonstrate significant reductions in frequency with outcomes improving over time, but only a minority of patients treated with these devices achieve seizure freedom (e.g., ~15% with RNS). Furthermore, no head-to-head device trials exist to establish relative superiority, so choice of a device is guided by patient-specific factors and by the strengths and limitations of each technology.

STATUS EPILEPTICUS

Status epilepticus refers to continuous seizures or repetitive, discrete seizures with impaired consciousness in the interictal period. Status epilepticus has numerous subtypes, including generalized convulsive status epilepticus (GCSE) (e.g., persistent, generalized electrographic seizures, coma, and tonic-clonic movements) and nonconvulsive status epilepticus (NCSE; e.g., persistent absence seizures or focal seizures with confusion or partially impaired consciousness, and minimal motor abnormalities). The duration of seizure activity sufficient to meet the definition of status epilepticus has traditionally been specified as 15–30 min. However, a more practical definition is to consider status epilepticus as a situation in which the duration of seizures prompts the acute use of anticonvulsant therapy. For GCSE, this is typically when seizures last beyond 5 min.

GCSE is an emergency and must be treated immediately, because cardiorespiratory dysfunction, hyperthermia, and metabolic derangements can develop as a consequence of prolonged seizures, and these

can lead to irreversible CNS injury. Furthermore, CNS injury can occur even when the patient is paralyzed with neuromuscular blockade but continues to have electrographic seizures. The most common causes of GCSE are anticonvulsant withdrawal or noncompliance, metabolic disturbances, drug toxicity, CNS infection, CNS tumors, refractory epilepsy, and head trauma.

GCSE is obvious when the patient is having overt seizures. However, after 30–45 min of uninterrupted seizures, the signs may become increasingly subtle. Patients may have mild clonic movements of only the fingers or fine, rapid movements of the eyes. There may be paroxysmal episodes of tachycardia, hypertension, and pupillary dilation. In such cases, the EEG may be the only method of establishing the diagnosis. Thus, if the patient stops having overt seizures, yet remains comatose, an EEG should be performed to rule out ongoing status epilepticus. This is obviously also essential when a patient with GCSE has been paralyzed with neuromuscular blockade in the process of protecting the airway.

The first steps in the management of a patient in GCSE are to attend to any acute cardiorespiratory problems or hyperthermia, perform a brief medical and neurologic examination, establish venous access, and send samples for laboratory studies to identify metabolic abnormalities. Anticonvulsant therapy should then begin without delay (Fig. 436-5).

The treatment of NCSE is thought to be less urgent than GCSE, because the ongoing seizures are not accompanied by the severe metabolic disturbances seen with GCSE. However, evidence suggests that NCSE, especially that caused by ongoing, focal seizure activity, is associated with cellular injury in the region of the seizure focus; therefore, this condition should be treated as promptly as possible using the general approach described for GCSE. Portable headband devices that provide a limited form of EEG can be rapidly applied without skilled technicians to help rule out NCSE in acute care settings.

BEYOND SEIZURES: OTHER MANAGEMENT ISSUES

EPILEPSY COMORBIDITIES

The adverse effects of epilepsy often go beyond clinical seizures. Many people with epilepsy feel completely normal between seizures and live highly successful and productive lives. However, a significant proportion of patients suffer from varying degrees of cognitive dysfunction, including psychiatric disease, and it has become increasingly clear that the network dysfunction underlying epilepsy can have effects well beyond the occurrence of seizures. For example, patients with seizures secondary to developmental abnormalities or acquired brain injury may have impaired cognitive function and other neurologic deficits due to abnormal brain structure. Frequent interictal EEG abnormalities are associated with subtle dysfunction of memory and attention. Patients with many seizures, especially those emanating from the temporal lobe, often note an impairment of short-term memory that may progress over time.

The psychiatric problems associated with epilepsy include depression, anxiety, and psychosis. This risk varies considerably depending on many factors, including the etiology, frequency, and severity of seizures and the patient's age and previous personal or family history of psychiatric disorder. Depression occurs in ~20–30% of patients, and the incidence of suicide is higher in people with epilepsy than in the general population. Depression should be treated through counseling and/or medication. The selective serotonin reuptake inhibitors (SSRIs) typically have minimal effect on seizures, whereas tricyclic antidepressants may lower the seizure threshold. Anxiety can be a seizure symptom, and anxious or psychotic behavior can occur during a postictal delirium. Postictal psychosis is a rare phenomenon that typically occurs after a period of increased seizure frequency. There is

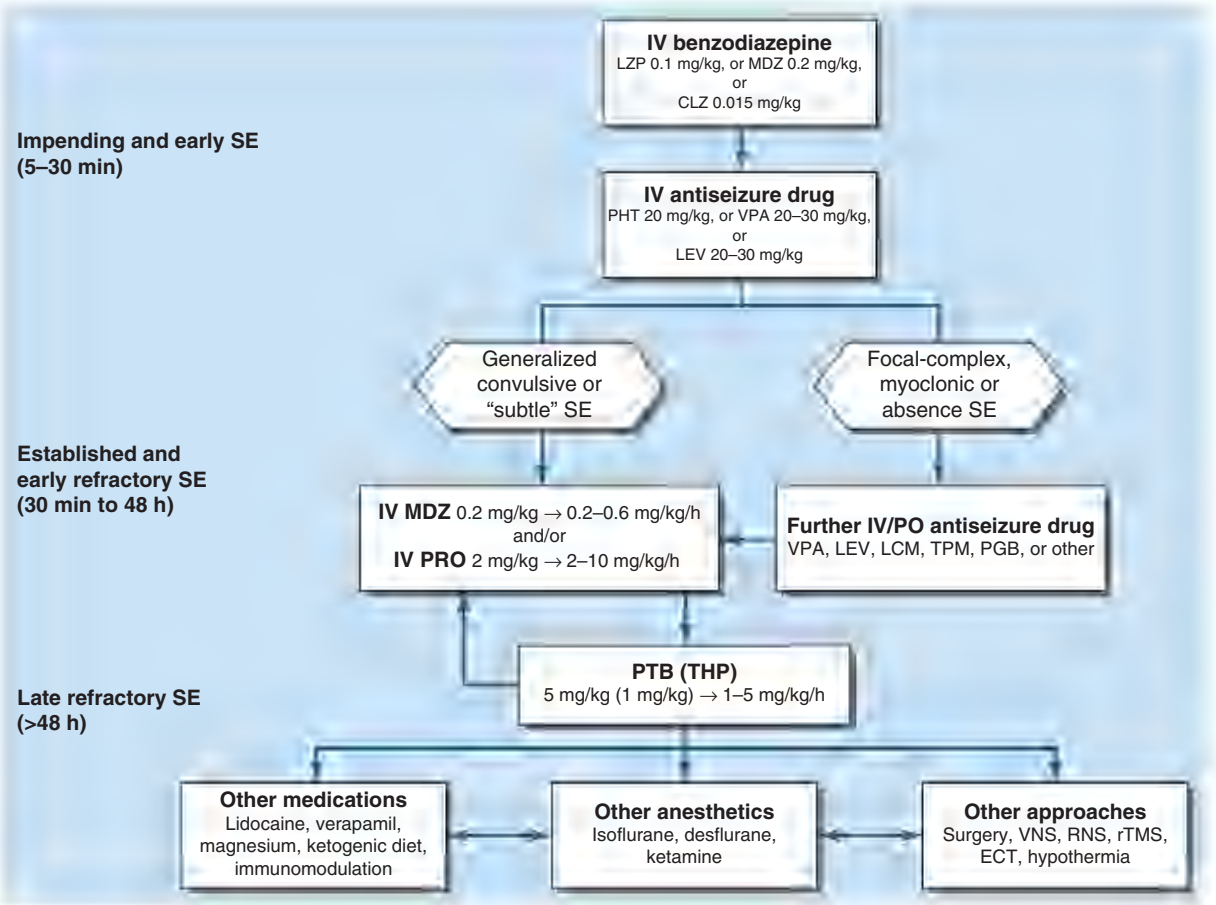

FIGURE 436-5 Pharmacologic treatment of generalized tonic-clonic status epilepticus (SE) in adults. CLZ, clonazepam; ECT, electroconvulsive therapy; LCM, lacosamide; LEV, levetiracetam; LZP, lorazepam; MDZ, midazolam; PGB, pregabalin; PHT, phenytoin or fosphenytoin; PRO, propofol; PTB, pentobarbital; RNS, responsive neurostimulation; rTMS, repetitive transcranial magnetic stimulation; THP, thiopental; TPM, topiramate; VNS, vagus nerve stimulation; VPA, valproic acid. *(Data from AO Rossetti, DH Lowenstein: Management of refractory status epilepticus in adults: still more questions than answers. Lancet Neurol 10:922, 2011.)*

usually a brief lucid interval lasting up to a week, followed by days to weeks of agitated, psychotic behavior. The psychosis usually resolves spontaneously but frequently will require short-term treatment with antipsychotic or anxiolytic medications.

MORTALITY OF EPILEPSY

People with epilepsy have a risk of death that is roughly two to three times greater than expected in a matched population without epilepsy. Most of the increased mortality is due to the underlying etiology of epilepsy (e.g., tumors or strokes in older adults). However, a significant number of patients die from accidents, status epilepticus, and a syndrome known as *sudden unexpected death in epilepsy* (SUDEP), which usually affects young people with convulsive seizures and tends to occur at night. The cause of SUDEP is unknown; it may result from brainstem-mediated effects of seizures on pulmonary, cardiac, and arousal functions. Genetic mutations may be the cause of both epilepsy and a cardiac conduction defect that gives rise to sudden death.

PSYCHOSOCIAL ISSUES

There continues to be a cultural stigma about epilepsy, although it is slowly declining in societies with effective health education programs. Many people with epilepsy harbor fear of progressive cognitive decline or dying during a seizure. These issues need to be carefully addressed by educating the patient about epilepsy and by ensuring that family members, teachers, fellow employees, and other associates are equally well informed. A useful source of educational material is the website *www.epilepsy.com*.

EMPLOYMENT, DRIVING, AND OTHER ACTIVITIES

Many patients with epilepsy face difficulty in obtaining or maintaining employment, even when their seizures are well controlled. Federal and state legislation is designed to prevent employers from discriminating against people with epilepsy, and patients should be encouraged to understand and claim their legal rights. Patients in these circumstances also benefit greatly from the assistance of health providers who act as strong patient advocates.

Loss of driving privileges is one of the most disruptive social consequences of epilepsy. Physicians should be very clear about local regulations concerning driving and epilepsy because the laws vary considerably among states and countries. In all cases, it is the physician's responsibility to warn patients of the danger imposed on themselves and others while driving if their seizures are uncontrolled (unless the seizures are not associated with impairment of consciousness or motor control). In general, most states in the United States allow patients to drive after a seizure-free interval (on or off medications) of 3–18 months.

Patients with incompletely controlled seizures must also contend with the risk of being in other situations where an impairment of consciousness or loss of motor control could lead to major injury or death. Thus, depending on the type and frequency of seizures, many patients need to be instructed to avoid working at heights or with machinery or to have someone close by for activities such as bathing and swimming.

The importance of quantifying seizures in people living with epilepsy has catalyzed a burgeoning industry of wearable sensors, such as wristwatches, that can detect seizures through noninvasive measurement of physiologic variables. Generally, non-EEG devices either have low sensitivity or high false alarm rate, but reliability is highest for detection of tonic-clonic seizures.

SPECIAL ISSUES RELATED TO WOMEN AND EPILEPSY

CATAMENIAL EPILEPSY

Some women experience a marked increase in seizure frequency around the time of menses. This is believed to be mediated by either the effects of estrogen and progesterone on neuronal excitability or changes in antiseizure drug levels due to altered protein binding or metabolism. Vulnerability to seizures is typically highest just before and during menses and during ovulation due to relatively high estrogen and low progesterone levels. Some women with epilepsy may benefit from increases in antiseizure drug dosages during menses. Natural progestins or intramuscular medroxyprogesterone may be of benefit to a subset of women.

PREGNANCY

Most women with epilepsy who become pregnant will have an uncomplicated gestation and deliver a normal baby. However, epilepsy poses some important risks to a pregnancy. Seizure frequency during pregnancy will remain unchanged in ~50% of women, increase in ~30%, and decrease in ~20%. Changes in seizure frequency are attributed to endocrine effects on the CNS, variations in antiseizure drug pharmacokinetics (such as acceleration of hepatic drug metabolism, increased renal blood flow, increased volume of distribution, or effects on plasma protein binding), or decreased antiseizure medication absorption due to nausea and vomiting. It is useful to see patients at frequent intervals during pregnancy and monitor serum antiseizure drug levels monthly; the risk of seizure exacerbation increases when serum levels decrease >35% from prepregnancy levels. Measurement of the unbound drug concentrations may be useful if there is an increase in seizure frequency or worsening of side effects of antiseizure drugs.

The overall incidence of fetal abnormalities in children born to mothers with epilepsy is 5–6%, compared to 2–3% in healthy women. Part of the higher incidence is due to teratogenic effects of antiseizure drugs, and the risk increases with the number of medications used (e.g., 10–20% risk of malformations with three drugs) and possibly with higher doses. A meta-analysis of published pregnancy registries and cohorts found that the most common malformations were defects in the cardiovascular and musculoskeletal system (1.4–1.8%). Valproic acid is strongly associated with an increased risk of adverse fetal outcomes (7–20%). Findings from a large pregnancy registry suggest that the newer antiseizure drugs are far safer than valproic acid.

Because the potential harm of uncontrolled convulsive seizures on the mother and fetus is considered greater than the teratogenic effects of antiseizure drugs, it is currently recommended that pregnant women be maintained on effective drug therapy. When possible, it seems prudent to have the patient on monotherapy at the lowest effective dose, especially during the first trimester. For some women, however, the type and frequency of their seizures may allow for them to safely wean off antiseizure drugs prior to conception. Patients should also take folate (1–4 mg/d), because the antifolate effects of anticonvulsants are thought to play a role in the development of neural tube defects, although the benefits of this treatment remain unproven in this setting.

Enzyme-inducing drugs such as phenytoin, carbamazepine, oxcarbazepine, topiramate, phenobarbital, and primidone cause a transient and reversible deficiency of vitamin K–dependent clotting factors in ~50% of newborn infants. Although neonatal hemorrhage is uncommon, the mother should be treated with oral vitamin K (20 mg/d, phylloquinone) in the last 2 weeks of pregnancy, and the infant should receive intramuscular vitamin K (1 mg) at birth.

CONTRACEPTION

Special care should be taken when prescribing antiseizure medications for women who are taking oral contraceptive agents. Drugs such as carbamazepine, phenytoin, phenobarbital, and topiramate can significantly decrease the efficacy of oral contraceptives via enzyme induction and other mechanisms. Patients should be advised to consider alternative forms of contraception, including intrauterine devices and other long-acting reversible contraceptives, or their oral contraceptive medications should be modified to offset the effects of the antiseizure medications. Estrogen-containing oral contraceptive agents induce glucuronidation of lamotrigine and can decrease lamotrigine serum levels by >50%.

BREAST-FEEDING

Antiseizure medications are excreted into breast milk, and the ratio of drug concentration in breast milk relative to serum ranges from ~5% (valproic acid) to 300% (levetiracetam). Given the overall benefits of breast-feeding and the lack of evidence for long-term harm to the infant by being exposed to antiseizure drugs, mothers with epilepsy should be encouraged to breast-feed unless there is evidence of drug effects on the infant, such as lethargy or poor feeding.